Unsupervised Learning Trade and Ahead

PGP DSBA - Guided Project Report

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12-8-2024

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1. Description:- Unsupervised Learning: Trade & Ahead Data Analysis:-

Context

The stock market has consistently proven to be a good place to invest in and save for the future. There are a lot of compelling reasons to invest in stocks. It can help in fighting inflation, create wealth, and also provides some tax benefits. Good steady returns on investments over a long period of time can also grow a lot more than seems possible. Also, thanks to the power of compound interest, the earlier one starts investing, the larger the corpus one can have for retirement. Overall, investing in stocks can help meet life's financial aspirations.

It is important to maintain a diversified portfolio when investing in stocks in order to maximise earnings under any market condition. Having a diversified portfolio tends to yield higher returns and face lower risk by tempering potential losses when the market is down. It is often easy to get lost in a sea of financial metrics to analyze while determining the worth of a stock, and doing the same for a multitude of stocks to identify the right picks for an individual can be a tedious task. By doing a cluster analysis, one can identify stocks that exhibit similar characteristics and ones which exhibit minimum correlation. This will help investors better analyze stocks across different market segments and help protect against risks that could make the portfolio vulnerable to losses.

Objective

Trade&Ahead is a financial consultancy firm who provide their customers with personalized investment strategies. They have hired you as a Data Scientist and provided you with data comprising stock price and some financial indicators for a few companies listed under the New York Stock Exchange. They have assigned you the tasks of analyzing the data, grouping the stocks based on the attributes provided, and sharing insights about the characteristics of each group.

2. Data Description

The data contains the different attributes of leads and their interaction details with Trade&Ahead investment strategies. The detailed data dictionary is given below.

2.1 Data Dictionary

- Ticker Symbol: An abbreviation used to uniquely identify publicly traded shares of a particular stock on a particular stock market
- Company: Name of the company
- GICS Sector: The specific economic sector assigned to a company by the Global Industry Classification Standard (GICS) that best defines its business operations
- GICS Sub Industry: The specific sub-industry group assigned to a company by the Global Industry Classification Standard (GICS) that best defines its business operations
- Current Price: Current stock price in dollars
- Price Change: Percentage change in the stock price in 13 weeks
- Volatility: Standard deviation of the stock price over the past 13 weeks
- ROE: A measure of financial performance calculated by dividing net income by shareholders' equity (shareholders' equity is equal to a company's assets minus its debt)

- Cash Ratio: The ratio of a company's total reserves of cash and cash equivalents to its total current liabilities
- Net Cash Flow: The difference between a company's cash inflows and outflows (in dollars)
- Net Income: Revenues minus expenses, interest, and taxes (in dollars)
- Earnings Per Share: Company's net profit divided by the number of common shares it has outstanding (in dollars)
- Estimated Shares Outstanding: Company's stock currently held by all its shareholders
- P/E Ratio: Ratio of the company's current stock price to the earnings per share
- P/B Ratio: Ratio of the company's stock price per share by its book value per share (book value of a company is the net difference between that company's total assets and total liabilities)

3. Overview of the Dataset

3.1 Checking the shape of the dataset:-

- The dataset contains information (15 attributes) about 340 rows.

3.2 Displaying few rows of the dataset:-

- The data frame has 15 columns as mentioned in the Data Dictionary.
- Data in each row corresponds to the leads and their interaction details with Trade & Ahead investment strategies.

	Ticker Symbol	Security	GICS Sector	GICS Sub Industry	Current Price	Price Change	Volatility	ROE	Cash Ratio	Net Cash Flow	Net Income	Earnings Per Share	Estimated Shares Outstanding	P/E Ratio	P/B Ratio
102	DVN	Devon Energy Corp.	Energy	Oil & Gas Exploration & Production	32.000000	-15.478079	2.923698	205	70	830000000	-14454000000	-35.55	4.065823e+08	93.089287	1.785616
125	FB	Facebook	Information Technology	Internet Software & Services	104.660004	16.224320	1.320606	8	958	592000000	3669000000	1.31	2.800763e+09	79.893133	5.884467
11	AIV	Apartment Investment & Mgmt	Real Estate	REITs	40.029999	7.578608	1.163334	15	47	21818000	248710000	1.52	1.636250e+08	26.335526	-1.269332
248	PG	Procter & Gamble	Consumer Staples	Personal Products	79.410004	10.660538	0.806056	17	129	160383000	636056000	3.28	4.913916e+08	24.070121	-2.256747
238	OXY	Occidental Petroleum	Energy	Oil & Gas Exploration & Production	67.610001	0.865287	1.589520	32	64	-588000000	-7829000000	-10.23	7.652981e+08	93.089287	3.345102
336	YUM	Yum! Brands Inc	Consumer Discretionary	Restaurants	52.516175	-8.698917	1.478877	142	27	159000000	1293000000	2.97	4.353535e+08	17.682214	-3.838260
112	EQT	EQT Corporation	Energy	Oil & Gas Exploration & Production	52.130001	-21.253771	2.364883	2	201	523803000	85171000	0.56	1.520911e+08	93.089287	9.567952
147	HAL	Halliburton Co.	Energy	Oil & Gas Equipment & Services	34.040001	-5.101751	1.966062	4	189	7786000000	-671000000	-0.79	8.493671e+08	93.089287	17.345857
89	DFS	Discover Financial Services	Financials	Consumer Finance	53.619999	3.653584	1.159897	20	99	2288000000	2297000000	5.14	4.468872e+08	10.431906	-0.375934
173	IVZ	Invesco Ltd.	Financials	Asset Management & Custody Banks	33.480000	7.067477	1.580839	12	67	412000000	968100000	2.26	4.283628e+08	14.814159	4.218620

Table 1 few rows of the dataset

3.3 Checking the data types of the columns for the dataset

- The column range index has 340 entries (i.e No.of rows = 340) which starts the number series from 0 to 339.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 340 entries, 0 to 339
Data columns (total 15 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Ticker Symbol                 340 non-null    object
 1   Security                      340 non-null    object
 2   GICS Sector                   340 non-null    object
 3   GICS Sub Industry             340 non-null    object
 4   Current Price                 340 non-null    float64
 5   Price Change                  340 non-null    float64
 6   Volatility                    340 non-null    float64
 7   ROE                           340 non-null    int64
 8   Cash Ratio                    340 non-null    int64
 9   Net Cash Flow                 340 non-null    int64
 10  Net Income                    340 non-null    int64
 11  Earnings Per Share            340 non-null    float64
 12  Estimated Shares Outstanding  340 non-null    float64
 13  P/E Ratio                     340 non-null    float64
 14  P/B Ratio                     340 non-null    float64
dtypes: float64(7), int64(4), object(4)
memory usage: 40.0+ KB
```

Table 2 data types

3.4 Checking for duplicate values

- There are 0 (i.e. zero) duplicate values.

3.5 Checking for missing values in the data

	0
Ticker Symbol	0
Security	0
GICS Sector	0
GICS Sub Industry	0
Current Price	0
Price Change	0
Volatility	0
ROE	0
Cash Ratio	0
Net Cash Flow	0
Net Income	0
Earnings Per Share	0
Estimated Shares Outstanding	0
P/E Ratio	0
P/B Ratio	0

dtype: int64

Table 3 missing values

3.6 Statistical summary of the dataset

Let's check the statistical summary of the data.

- This table reflects the statistical summary of the data contains - Count, Mean, Std (Standard deviation), Min (Minimum), Max (Maximum), and Percentages (25%, 50%, 75%).

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Ticker Symbol	340	340	AAL	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Security	340	340	American Airlines Group	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
GICS Sector	340	11	Industrials	53	NaN	NaN	NaN	NaN	NaN	NaN	NaN
GICS Sub Industry	340	104	Oil & Gas Exploration & Production	16	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Current Price	340.0	NaN	NaN	NaN	80.862345	98.055086	4.5	38.555	59.705	92.880001	1274.949951
Price Change	340.0	NaN	NaN	NaN	4.078194	12.006338	-47.129693	-0.939484	4.819505	10.695493	55.051683
Volatility	340.0	NaN	NaN	NaN	1.525976	0.591798	0.733163	1.134878	1.385593	1.695549	4.580042
ROE	340.0	NaN	NaN	NaN	39.597059	96.547538	1.0	9.75	15.0	27.0	917.0
Cash Ratio	340.0	NaN	NaN	NaN	70.023529	90.421331	0.0	18.0	47.0	99.0	958.0
Net Cash Flow	340.0	NaN	NaN	NaN	55537620.588235	1946365312.175789	-11208000000.0	-193906500.0	2098000.0	169810750.0	20764000000.0
Net Income	340.0	NaN	NaN	NaN	1494384602.941176	3940150279.327937	-23528000000.0	352301250.0	707336000.0	1899000000.0	24442000000.0
Earnings Per Share	340.0	NaN	NaN	NaN	2.776662	6.587779	-61.2	1.5575	2.895	4.62	50.09
Estimated Shares Outstanding	340.0	NaN	NaN	NaN	577028337.75403	845849595.417695	27672156.86	158848216.1	309675137.8	573117457.325	6159292035.0
P/E Ratio	340.0	NaN	NaN	NaN	32.612563	44.348731	2.935451	15.044653	20.819876	31.764755	528.039074
P/B Ratio	340.0	NaN	NaN	NaN	-1.718249	13.966912	-76.119077	-4.352056	-1.06717	3.917066	129.064585

Table 4 Statistical summary

- Count is 340 rows for all the attributes in the data set.
- Unique:- Ticker symbol - 340, Security - 340, GICS Sector – 11, GICS Sub Industry -104 has unique values and other columns do not have any unique values.
- Max: - Earning Per Share has maximum price of 50.09 in dollars.
- Min: - current price has minimum of 4.5 in dollars

Exploratory Data Analysis

- EDA has a key role in data analysis for the data science. It includes different types of analysis like Univariate analysis & Bivariate analysis which are explained in the below points.

4. Univariate Analysis:-

- It means one variable analysis with count. It refers to the observations of different variables.

4.1 Current Price

- It is to create a histogram_boxplot for 'Current Price'.

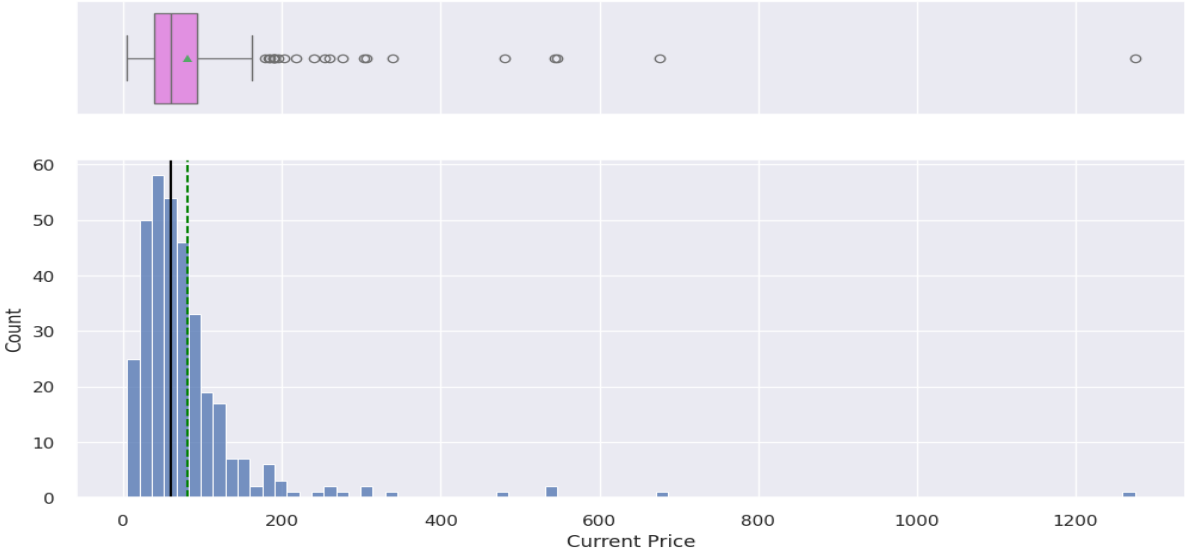

Figure 1 Current Price

- The current price observations are approximately 80.86 (Approximately) on average.
- It has many outliers on the right-hand side and it has the right side skewed due to outliers on the right side.
- The median is 59.70.

4.2 Price Change

- It is to create a histogram_boxplot for 'Price Change'.

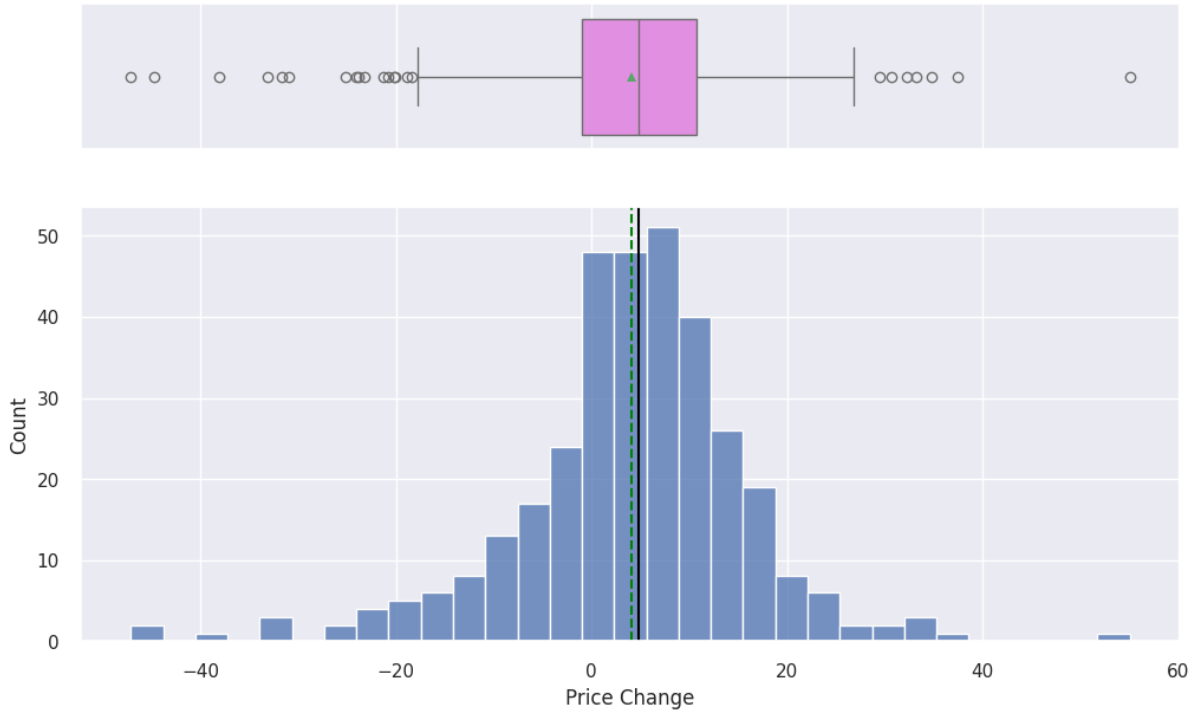

Figure 2 Price Change

- The Price Change observations are approximately 4.08 (Approximately) on average.
- It has many outliers on the left-hand side and 7 outliers on the right-hand side. It has slightly the left side skewed.
- The median is 4.81 It is close to normal distribution.

4.3 Volatility

- It is to create a histogram_boxplot for 'Volatility'.

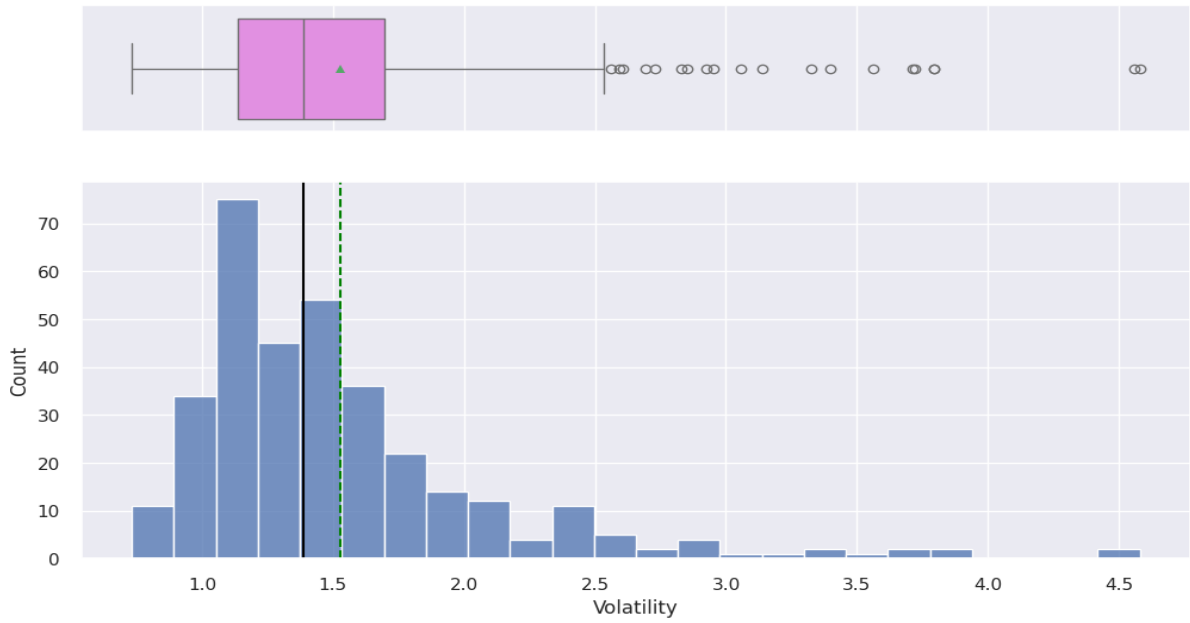

Figure 3 Volatility

- The Volatility observations are approximately 1.52 (Approximately) on average.

- It has many outliers on the right-hand side and it has the right side skewed.
- The median is 1.38.

4.4 ROE

- It is to create a histogram_boxplot for 'ROE'.

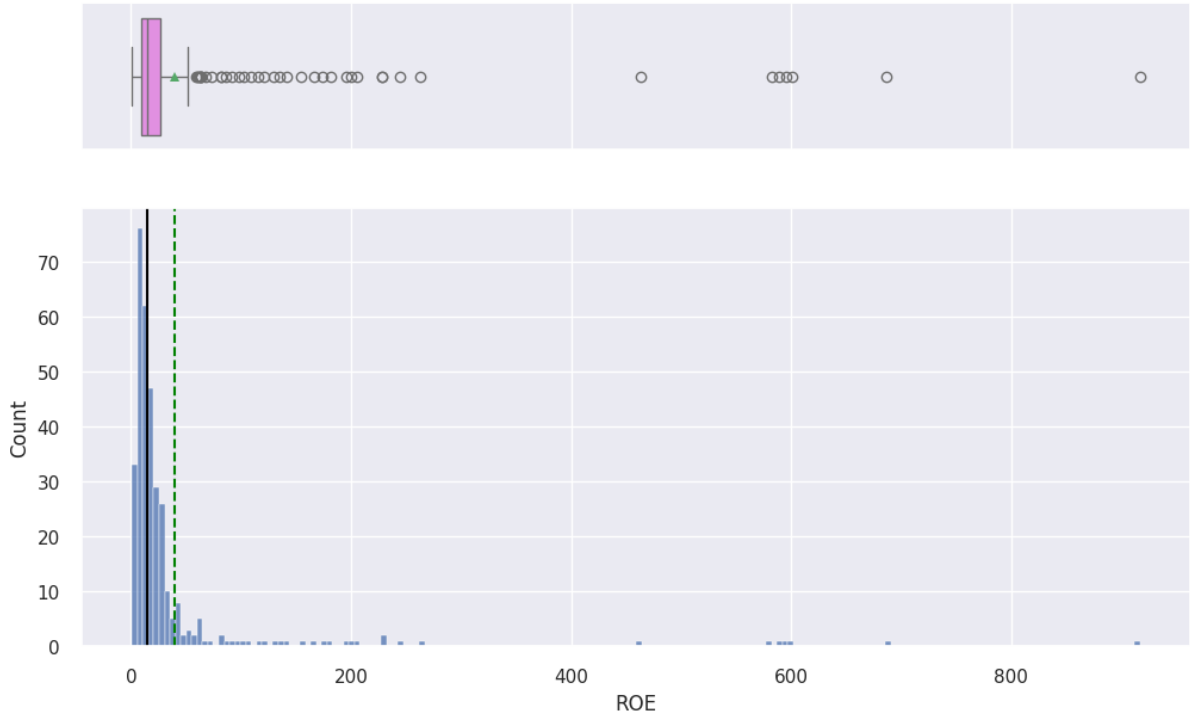

Figure 4 ROE

- The ROE observations are approximately 40 (Approximately) on average.
- It has too many outliers on the right-hand side and it has the right side skewed due to many outliers on the right side.
- The median is 15.

4.5 Cash Ratio

- It is to create a histogram_boxplot for 'Cash Ratio'.

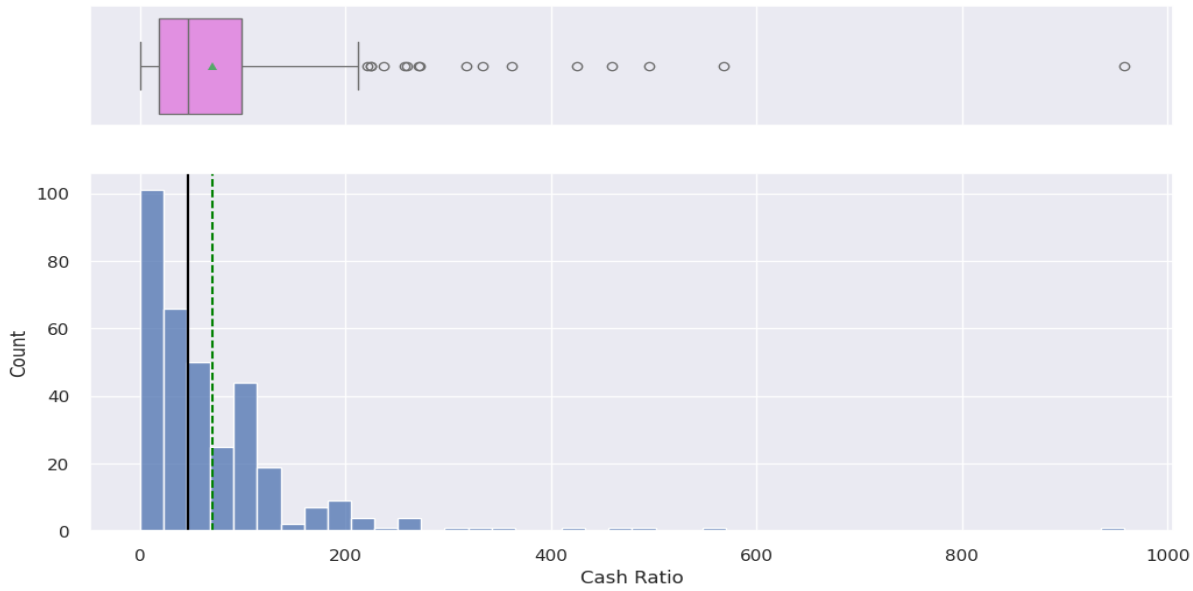

Figure 5 Cash Ratio

- The Cash Ratio observations are approximately 70.02 (Approximately) on average.
- It has many outliers on the right-hand side and it has the right side skewed.
- The median is 47. It is not close to normal distribution.

4.6 Net Cash Flow

- It is to create a histogram_boxplot for 'Net Cash Flow'.

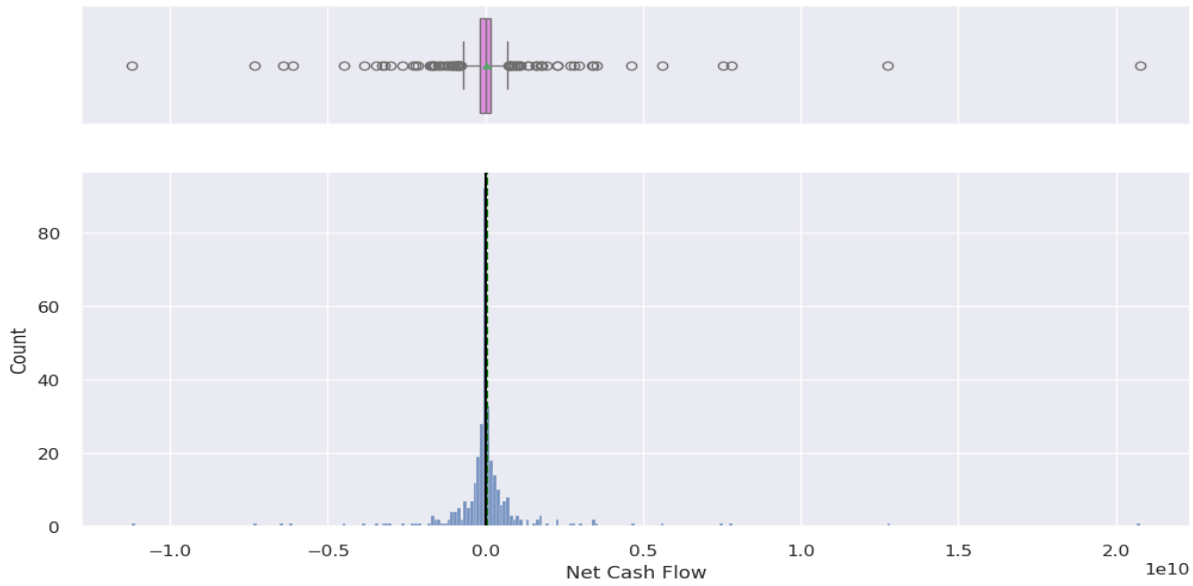

Figure 6 Net Cash Flow

- The Net Cash Flow observations are approximately 0.0055 (Approximately) on average.
- It has many outliers on both the right and left sides. The right side is skewed due to the greater number of outliers.
- The median is 0.0002. It is close to normal distribution.

4.7 Net Income

- It is to create a histogram_boxplot for 'Net Income'.

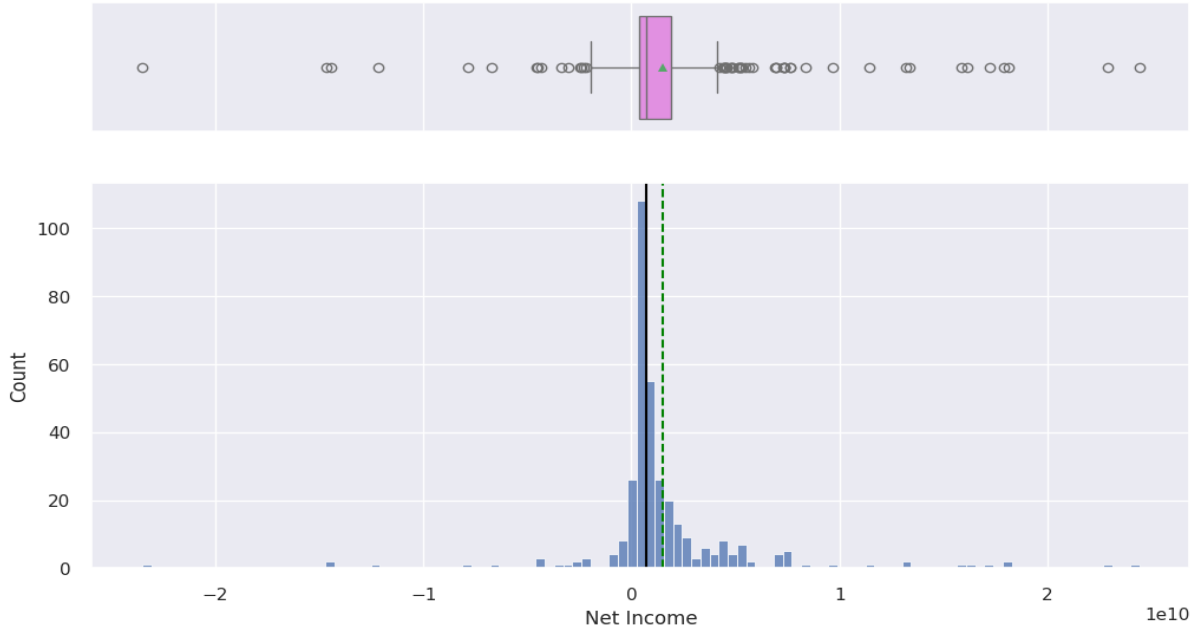

Figure 7 Net Income

- The Net Income observations are approximately 0.149 (Approximately) on average.
- It has many outliers on both the right and left sides. It has no skewness.
- The median is 0.07 (Approximately). It is close to normal distribution.

4.8 Earnings Per Share

- It is to create a histogram_boxplot for 'Earnings Per Share'.

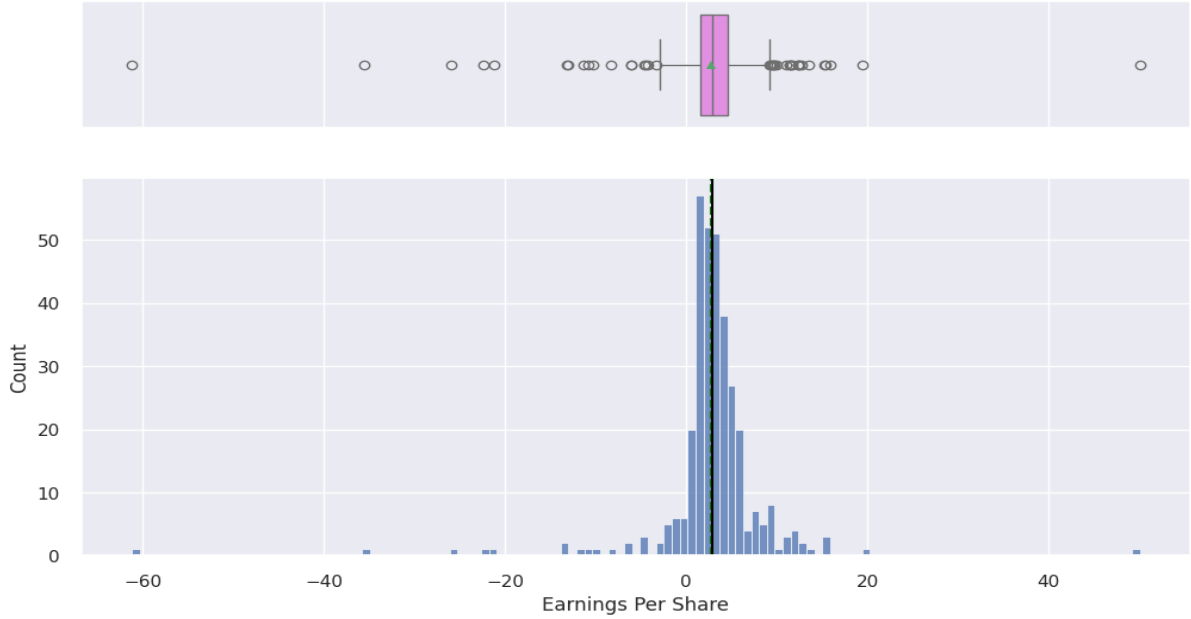

Figure 8 Earnings Per Share

- The Earnings Per Share observations are approximately 2.77 (Approximately) on average.
- It has many outliers on both the right and left sides. The left side is skewed due to the greater number of outliers.
- The median is 2.89 (Approximately). It is close to normal distribution.

4.9 Estimated Shares Outstanding

- It is to create a histogram_boxplot for 'Estimated Shares Outstanding'.

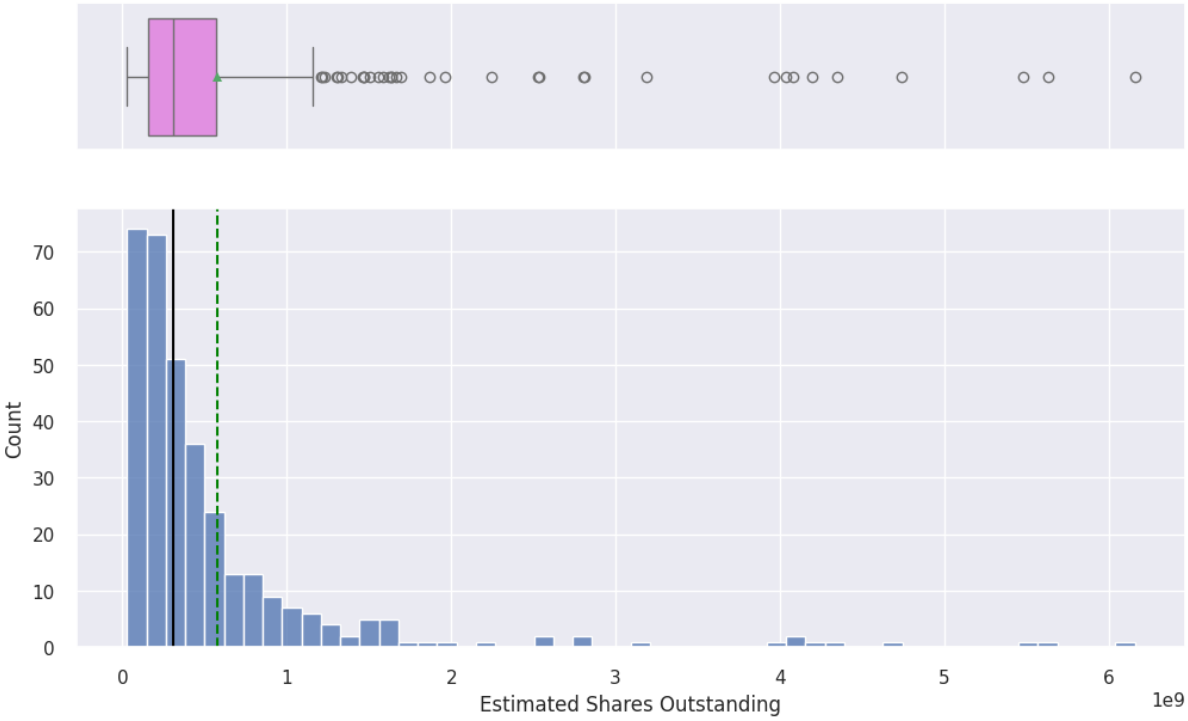

Figure 9 Estimated Shares Outstanding

- The Estimated Shares of Outstanding observations are approximately 0.577 (Approximately) on average.
- It has many outliers on both the right-hand side. It has skewed to the right side.
- The median is 0.309 (Approximately). It is not close to normal distribution.

4.10 P/E Ratio

- It is to create a histogram_boxplot for 'P/E Ratio'.

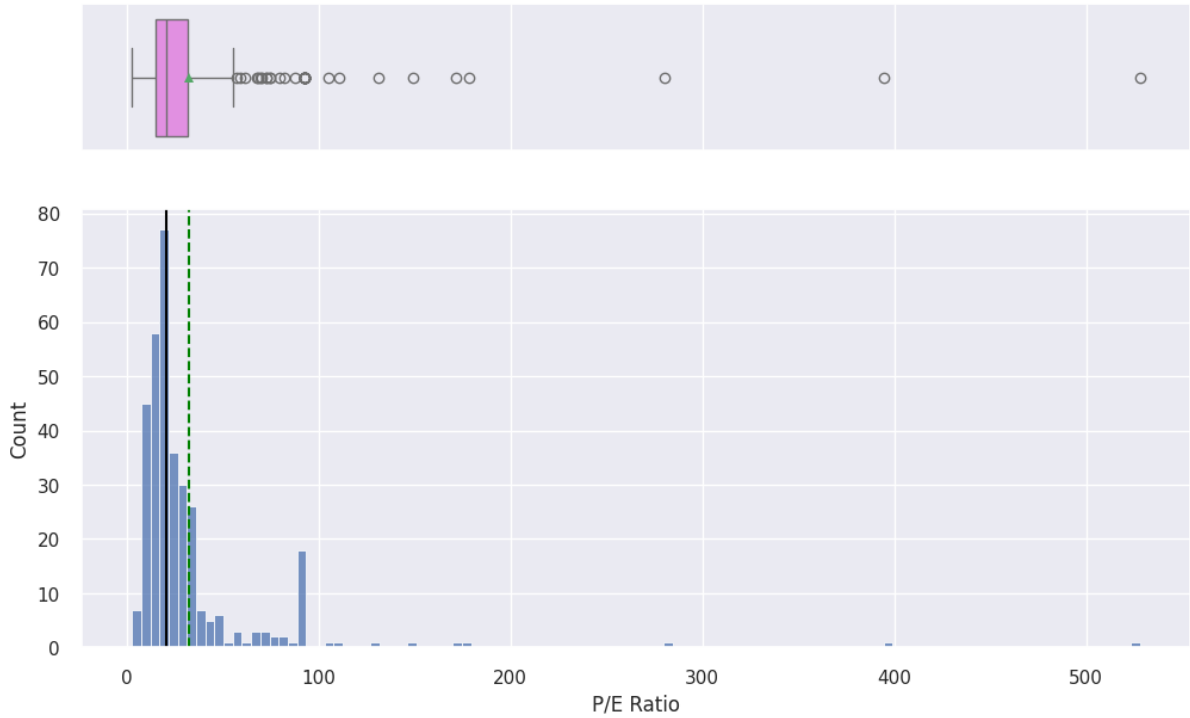

Figure 10 P/E Ratio

- The P/E Ratio observations are approximately 32.61 (Approximately) on average.
- It has many outliers on the right-hand side and it has the right side skewed.
- The median is 20.81 (Approximately). It is close to normal distribution.

4.11 P/B Ratio

- It is to create a histogram_boxplot for 'P/B Ratio'.

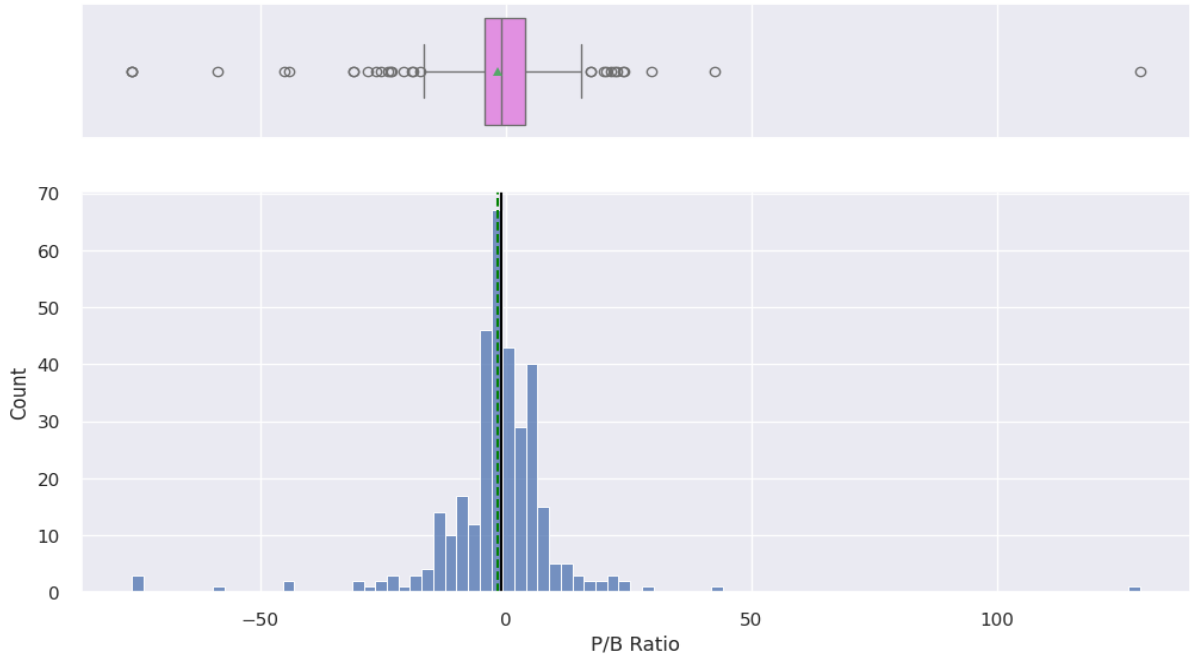

Figure 11 P/B Ratio

- The P/B Ratio observations are approximately -1.71 (Approximately) on average.
- It has many outliers on both the right-hand and left-hand sides. It has a right side skewed due to one big outlier showing on the right side.
- The median is -1.06. It is close to normal distribution.

5. Function to create labeled barplots

- It is to create a labeled barplots for 'GICS Sector wise'.

5.1 GICS Sector

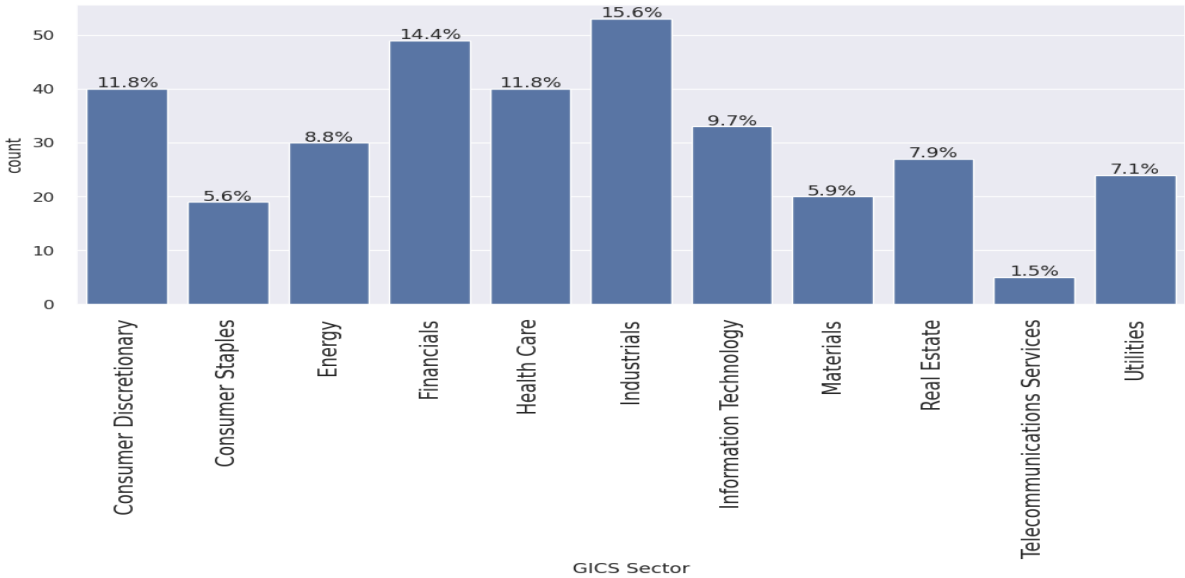

Figure 12 GICS Sector

- The industrial sector has the highest business operations with 15% across all GICS sectors followed by the Financials sector with 14.4%.
- Telecommunications Services occupied 1.5% which is the lowest.

5.2 GICS Sub Industry

- It is to create a labelled barplots for 'GICS Sub Industry wise'.

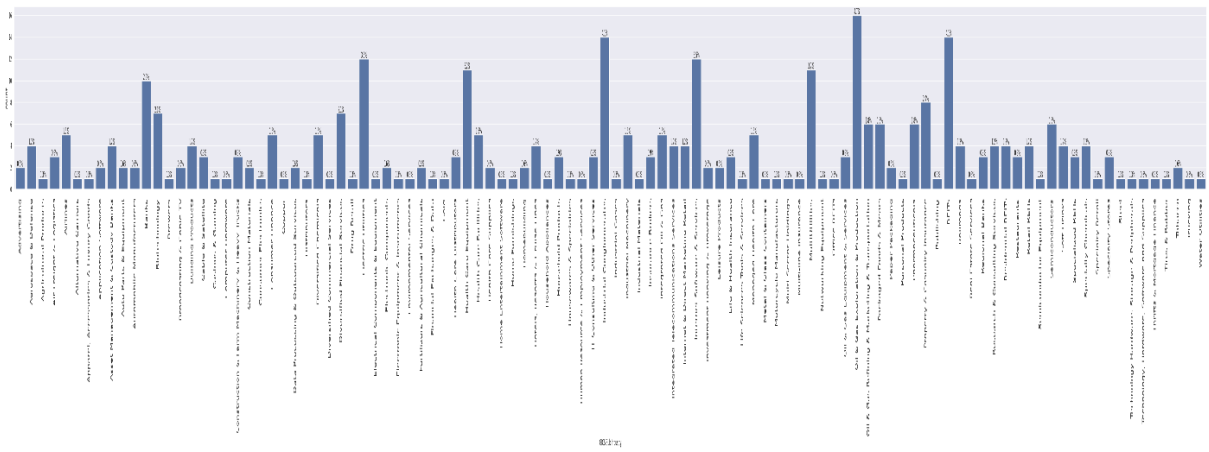

Figure 13 GICS Sub Industry

6. Bivariate Analysis

Let's see the attributes that have a strong correlation with each other.

6.1 Correlation Check:-

We used a Heatmap to view the correlation between parameters.

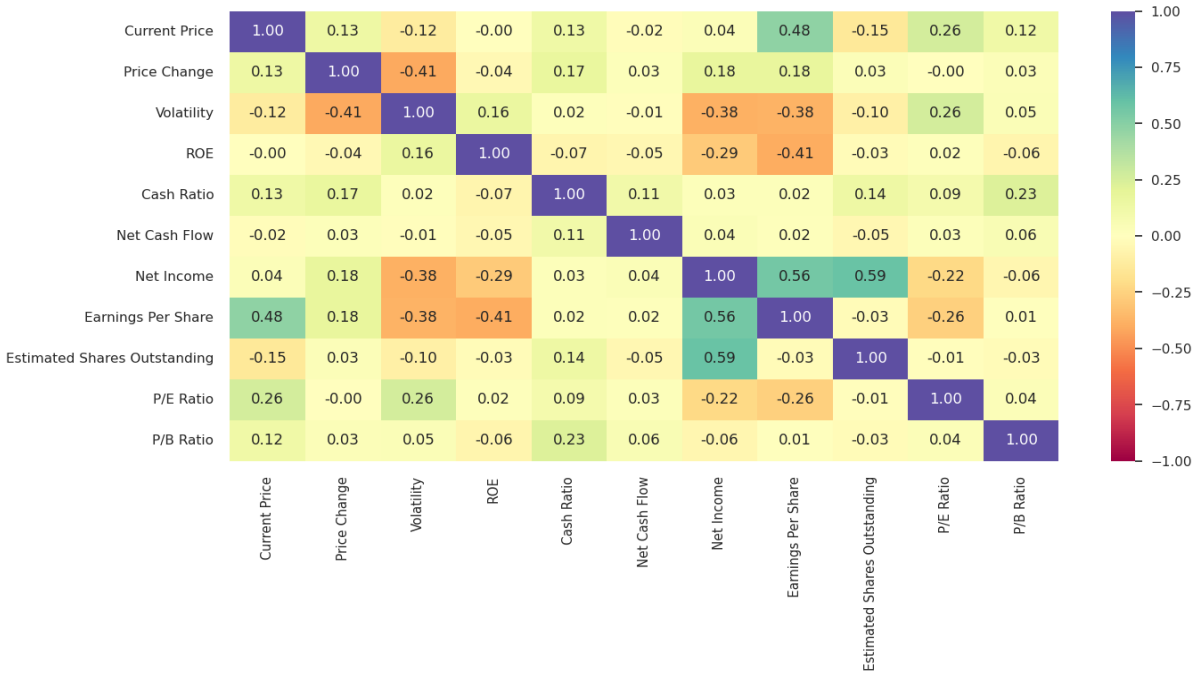

Figure 14 Correlation Check Heatmap

- Net Income and Estimated Shares Outstanding have a strong positive correlation (i.e 0.59)
- ROE has a strong adverse correlation (i.e -0.54) with Earnings Per Share.
- P/E Ratio has zero (i.e -0.00) correlation with Price Change.

6.2 Let's check the stocks of which economic sector have seen the maximum price increase on average.

Figure 15 stocks of which the economic sector has seen the maximum price increase on average

- The stocks in the Health Care sector have seen the maximum price increase on average. So it is the highest price change.
- The Consumer Staples sector occupies the second place for the price increase.

6.3 Cash ratio provides a measure of a company's ability to cover its short-term obligations using only cash and cash equivalents. Let's see how the average cash ratio varies across economic sectors.

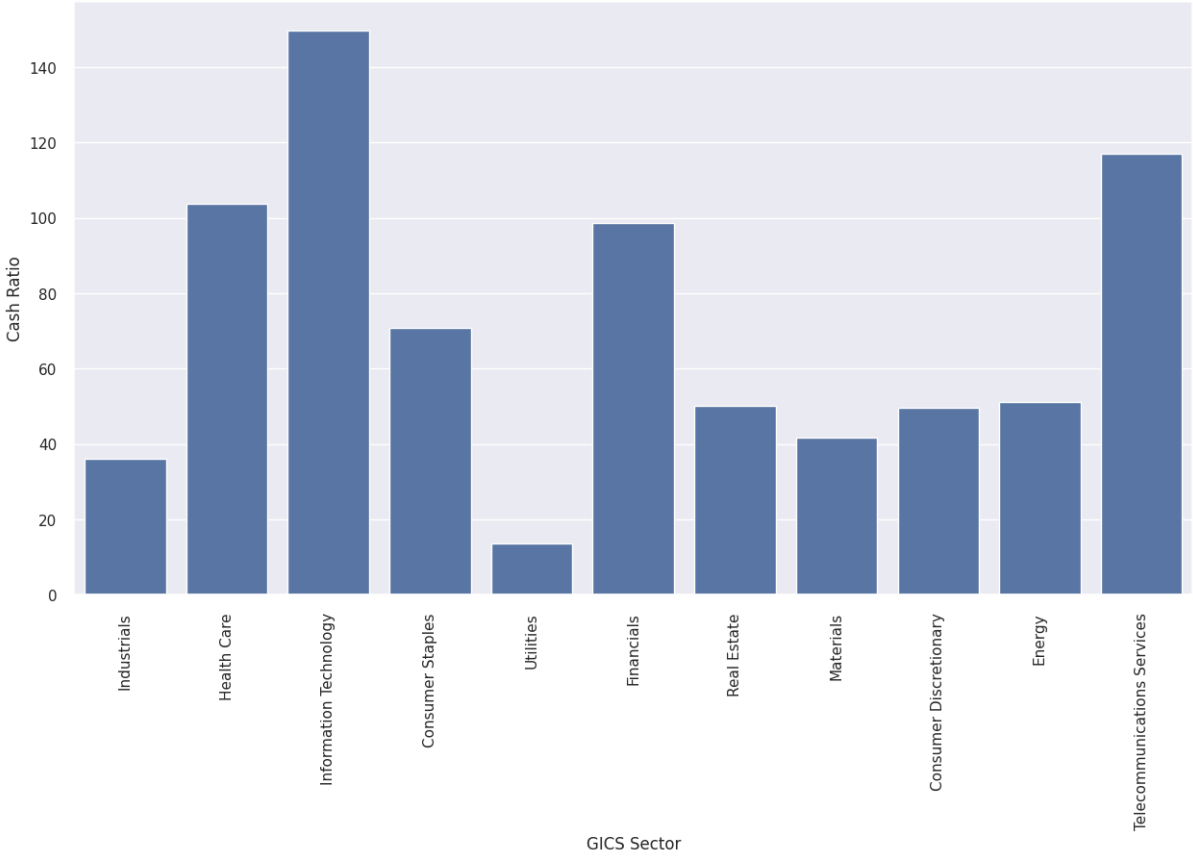

Figure 16 average cash ratio varies across economic sectors

- Information Technology has the highest Cash Ratio and Utilities are the least.
- Energy and Consumer Discretionary sectors have no variance across economic sectors.

6.4 P/E ratios can help determine the relative value of a company's shares as they signify the amount of money an investor is willing to invest in a single share of a company per dollar of its earnings. Let's see how the P/E ratio varies, on average, across economic sectors.

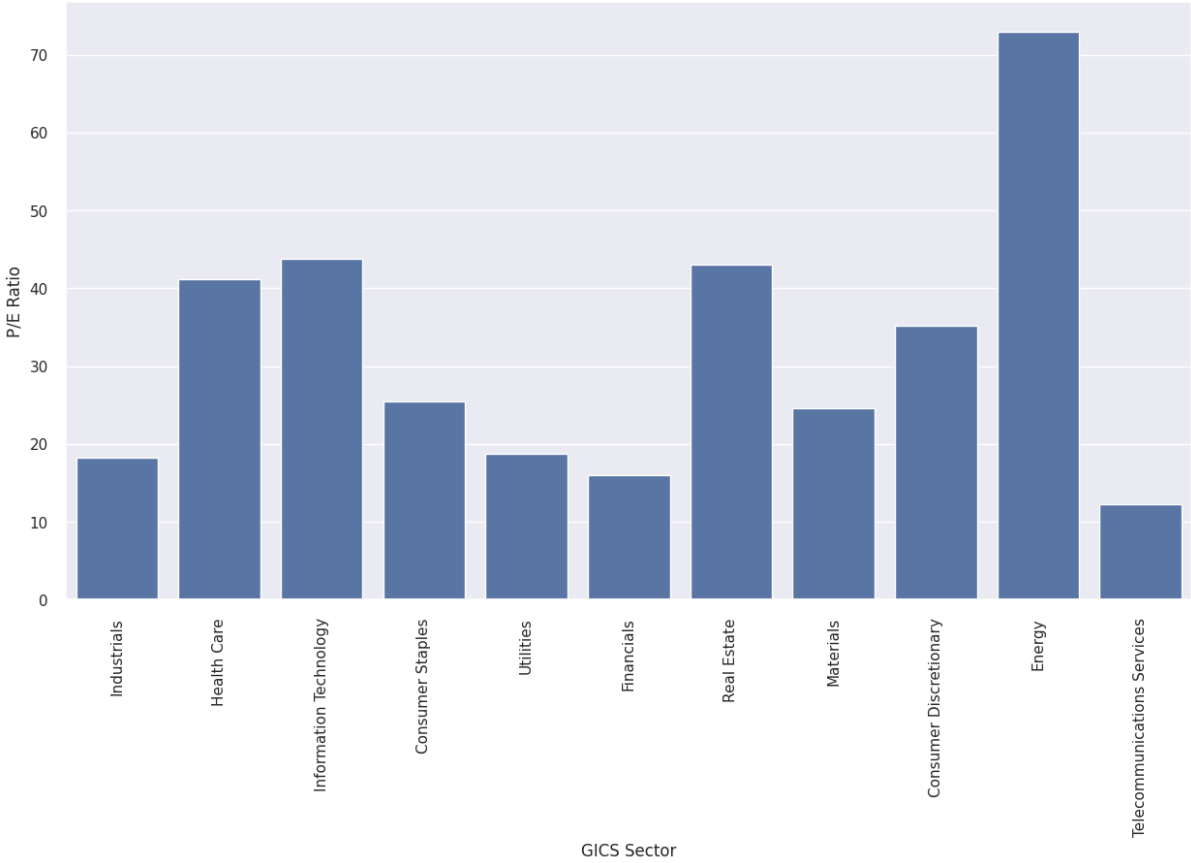

Figure 17 P/E ratio varies, on average, across economic sectors

- Energy has the highest P/E Ratio and Telecommunications Services shows the least place.
- Real Estate and Information Technology are very close with minimal variance.

6.5 Volatility accounts for the fluctuation in the stock price. A stock with high volatility will witness sharper price changes, making it a riskier investment. Let's see how volatility varies, on average, across economic sectors.

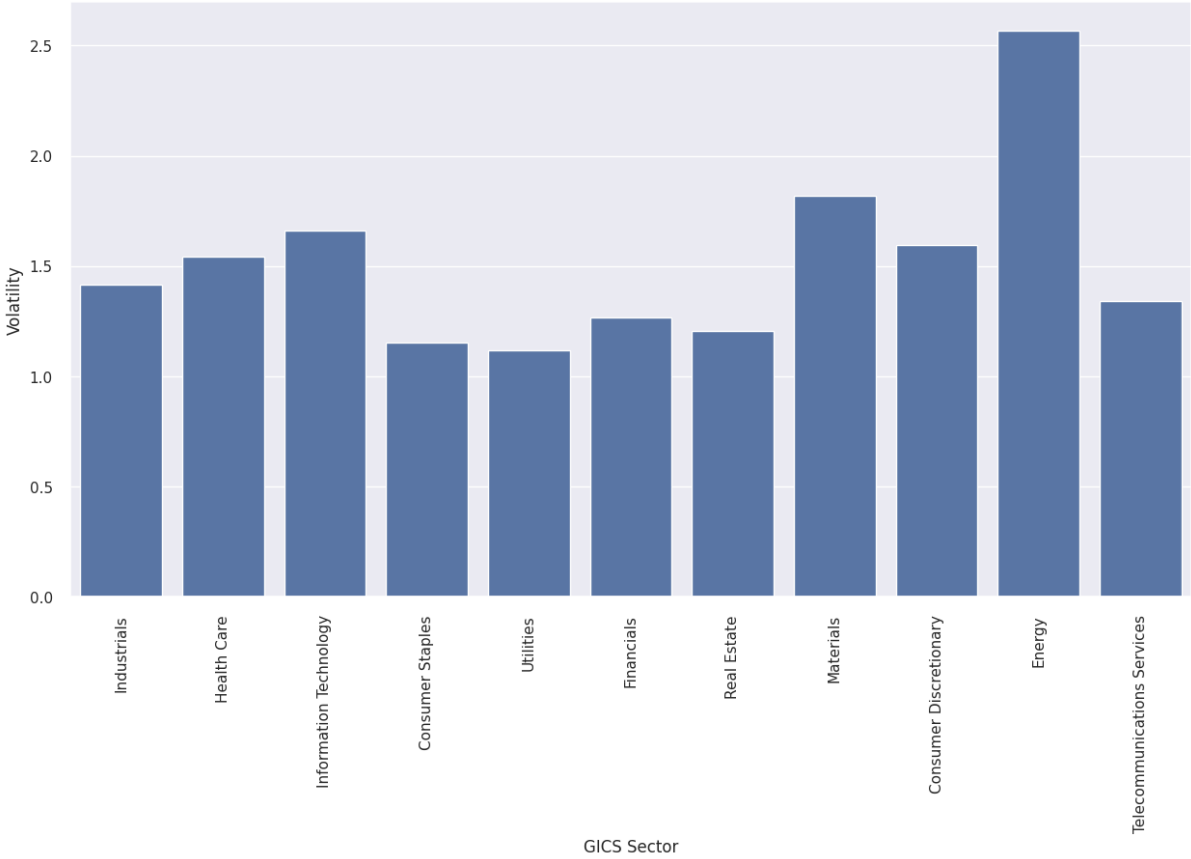

Figure 18 volatility varies, on average, across economic sectors

- Energy has highest volatility accounts for the fluctuation in the stock price.
- Utilities have the least volatility accounts for the fluctuation in the stock price.
- Energy, Materials, Consumer Discretionary, Information Technology, Health Care, and Industries have high volatility so these are risky investments.
- Utilities, Consumer Staples, Real estate, and Real Estate have less volatility accounts for the fluctuation in the stock price so these are risk-less investments.

7. Data Preprocessing

7.1 Outlier Check: –

- Let's plot the boxplots of all numerical columns to check for outliers.

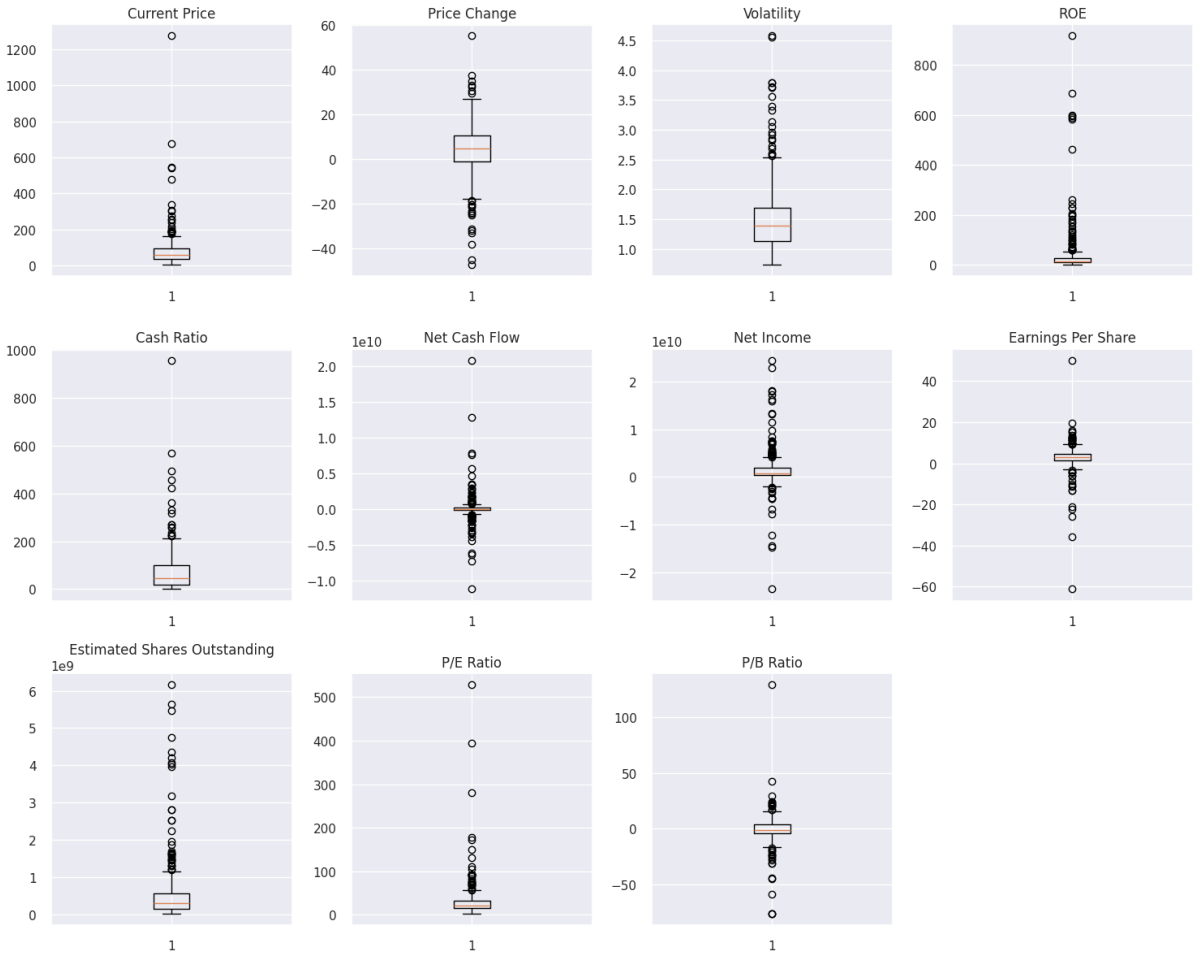

Figure 19 Outlier Check

7.2 Scaling

- We will scale the data before we proceed with clustering.
- It needs to create a data frame of the scaled data

8. K-means Clustering

8.1 Checking Elbow Plot

```
Number of Clusters: 1   Average Distortion: 2.5425069919221697
Number of Clusters: 2   Average Distortion: 2.3862098789299604
Number of Clusters: 3   Average Distortion: 2.33620927590848
Number of Clusters: 4   Average Distortion: 2.219050563833442
Number of Clusters: 5   Average Distortion: 2.133404401901685
Number of Clusters: 6   Average Distortion: 2.081503686093715
Number of Clusters: 7   Average Distortion: 2.0045413402786814
Number of Clusters: 8   Average Distortion: 1.9864237824874411
Number of Clusters: 9   Average Distortion: 1.956222103389025
Number of Clusters: 10  Average Distortion: 1.9360473996664198
Number of Clusters: 11  Average Distortion: 1.8615942883461607
Number of Clusters: 12  Average Distortion: 1.8219574388532505
Number of Clusters: 13  Average Distortion: 1.7936924742607907
Number of Clusters: 14  Average Distortion: 1.7567842179093438
```

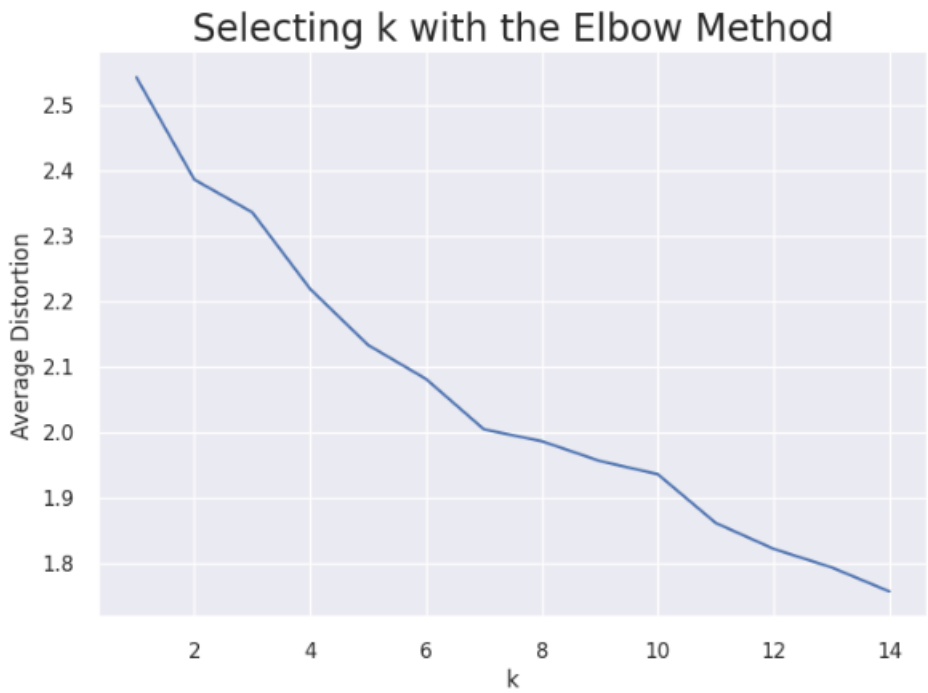

Figure 20 Checking Elbow Plot

Let's do further analysis to determine the optimal value of k

8.2 Data to the visualizer

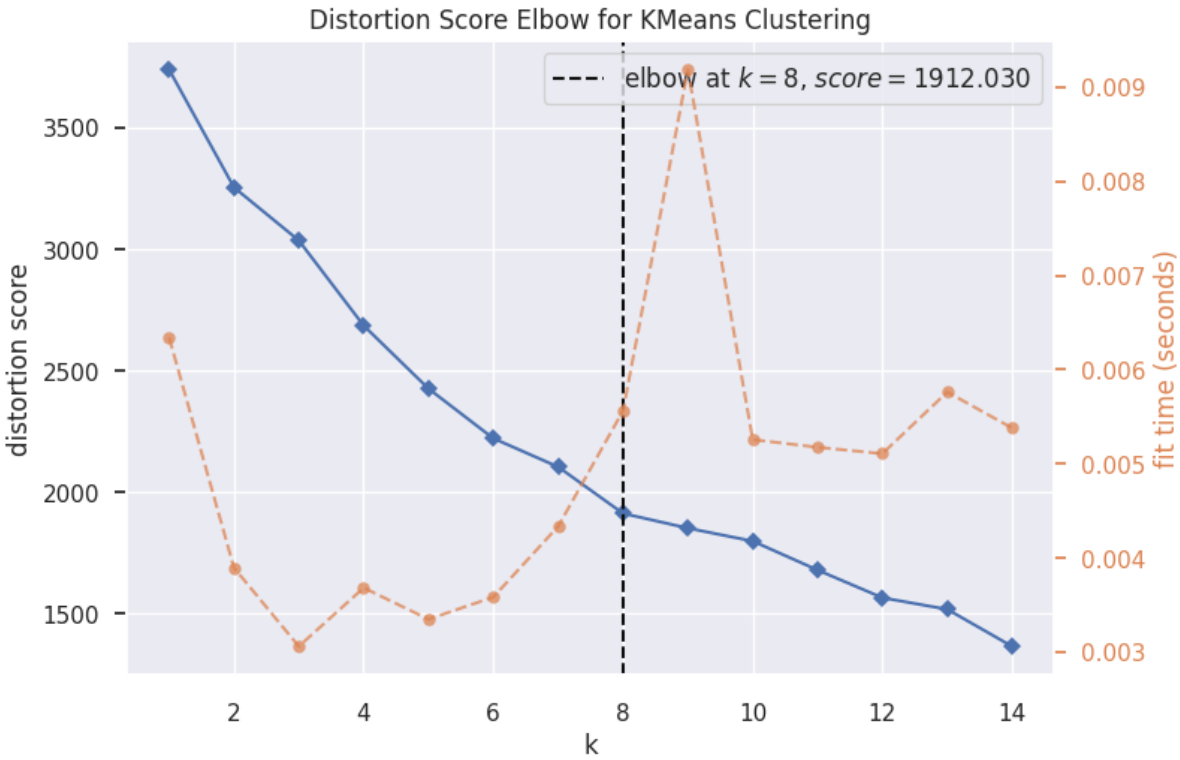

Figure 21 Data to the visualizer

<Axes: title={'center': 'Distortion Score Elbow for KMeans Clustering'}, xlabel='k', ylabel='distortion score'>

8.3 Let's check the silhouette scores

```
For n_clusters = 2, the silhouette score is 0.45335782729503565)
For n_clusters = 3, the silhouette score is 0.40374060030338865)
For n_clusters = 4, the silhouette score is 0.4246430808437099)
For n_clusters = 5, the silhouette score is 0.4381539778147092)
For n_clusters = 6, the silhouette score is 0.40869599703024256)
For n_clusters = 7, the silhouette score is 0.1207450219233897)
For n_clusters = 8, the silhouette score is 0.3693991650696542)
For n_clusters = 9, the silhouette score is 0.35185096182499204)
For n_clusters = 10, the silhouette score is 0.32950073703610283)
For n_clusters = 11, the silhouette score is 0.1486586842527321)
For n_clusters = 12, the silhouette score is 0.15784241071085106)
For n_clusters = 13, the silhouette score is 0.15646997458716602)
For n_clusters = 14, the silhouette score is 0.16253506827999134)
```

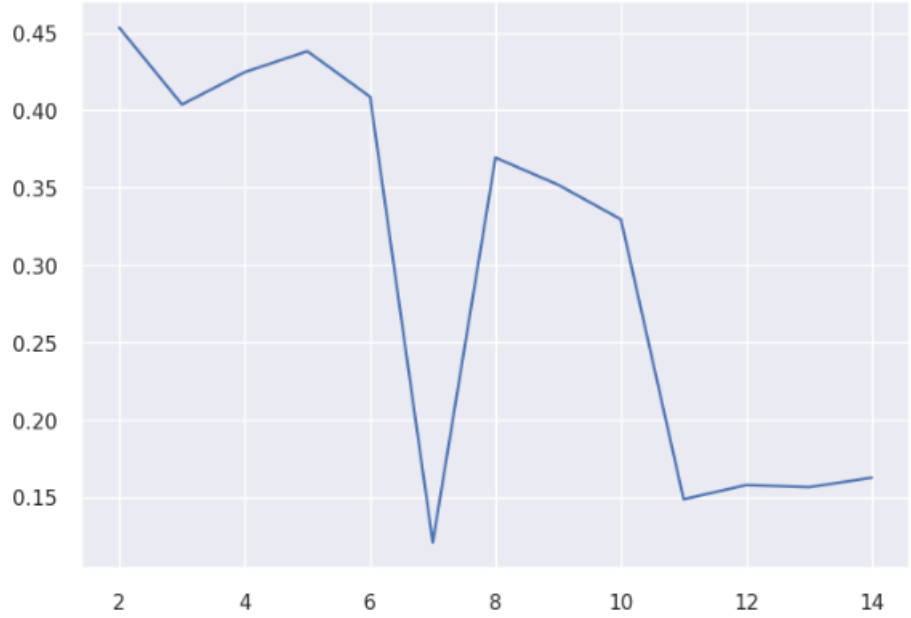

Figure 22 silhouette scores

Silhouette score Elbow

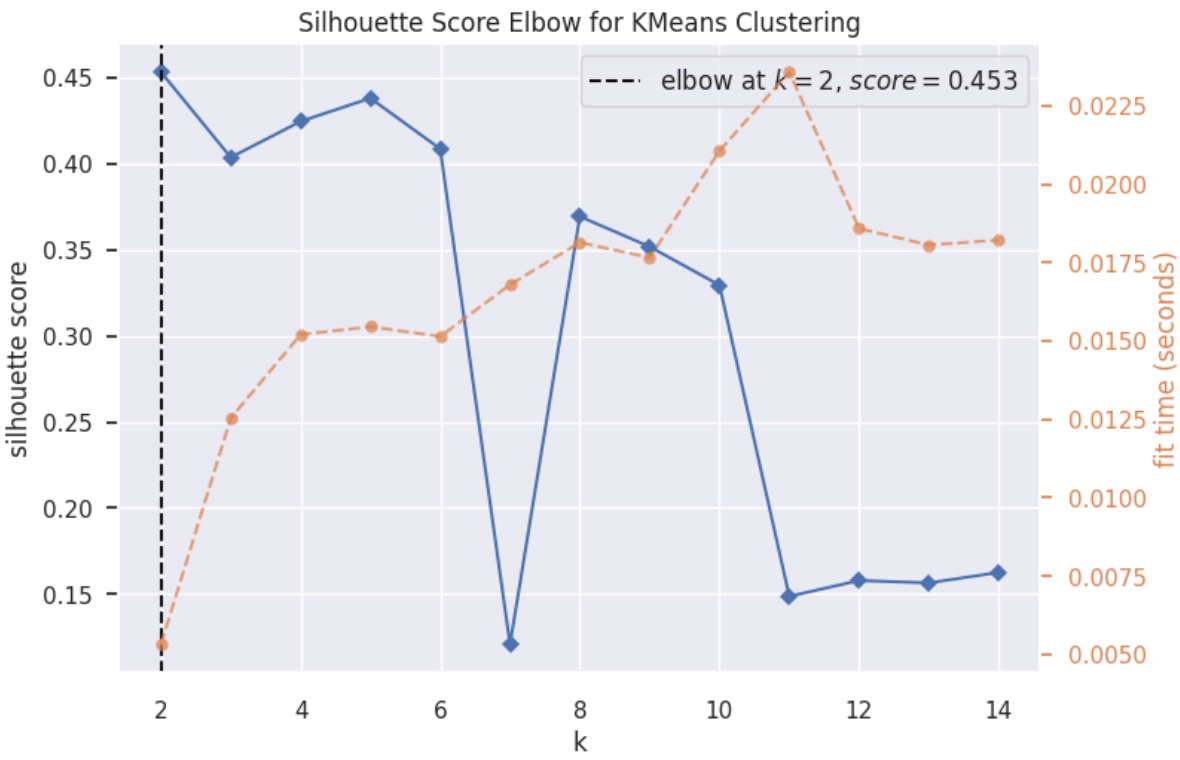

Figure 23 Silhouette score Elbow

<Axes: title={'center': 'Silhouette Score Elbow for KMeans Clustering'}, xlabel='k', ylabel='silhouette score'>

From the silhouette scores, it seems that 4 is a good value for k.

8.4 Silhouette Plot

Finding optimal no. of clusters with silhouette coefficients

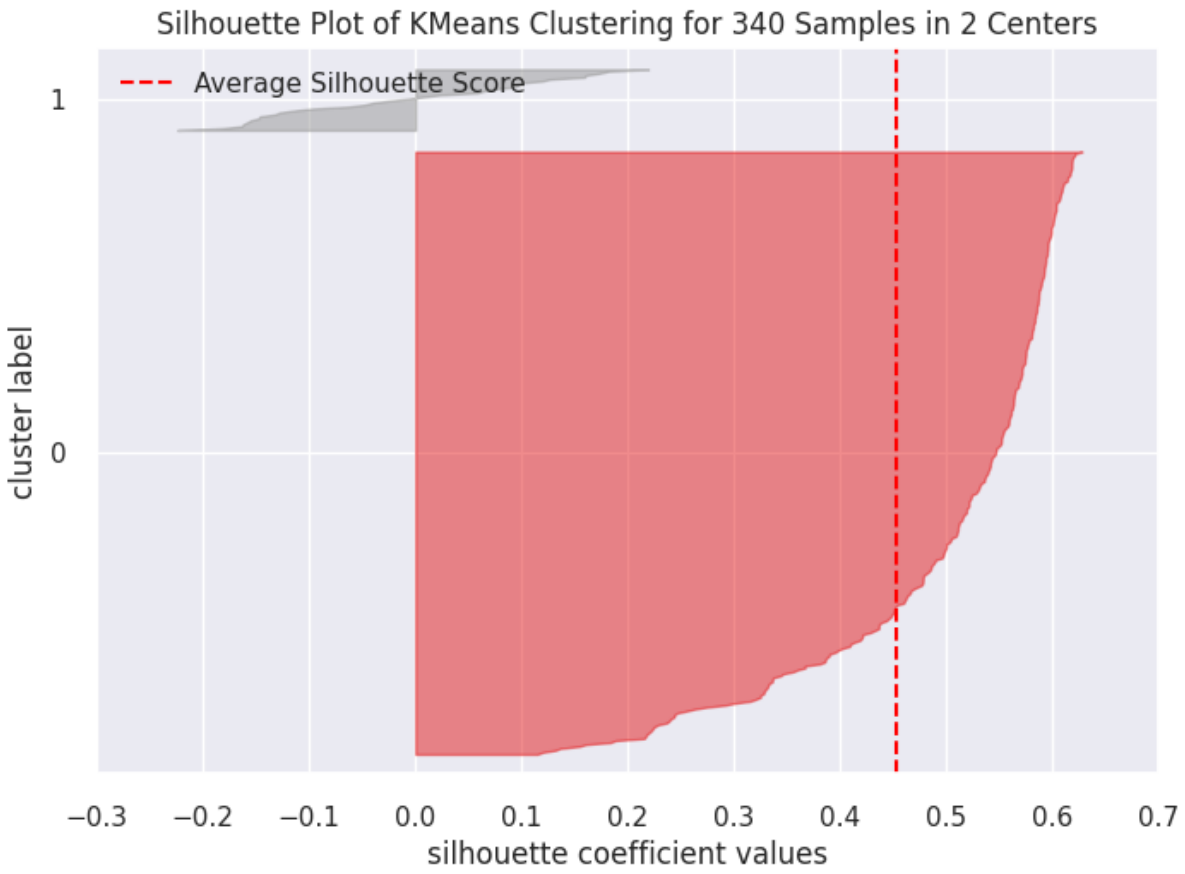

Figure 24 silhouette coefficients

- Axes: title={'center': 'Silhouette Plot of KMeans Clustering for 340 Samples in 2 Centers'}, xlabel='silhouette coefficient values', ylabel='cluster label'
- Cluster label 0 is more than average so it is Optimal.

8.5 Creating Final Model

final K-means model

```
KMeans
KMeans(n_clusters=2, random_state=1)
```

Figure 25 K-means model

8.6 Cluster Profiling

K _ Group by the cluster labels

KM_segments	Current Price	Price Change	Volatility	ROE	Cash Ratio	Net Cash Flow	Net Income	Earnings Per Share	Estimated Shares Outstanding	P/E Ratio	P/B Ratio	count_in_each_segment
0	82.111257	5.570491	1.394712	31.782468	70.564935	70043230.519481	1986009597.402597	3.889886	584677860.238409	24.668434	-1.778604	308
1	68.841564	-10.285163	2.789397	114.812500	64.812500	-84078875.000000	-3237505968.750000	-7.938125	503401683.841875	109.074811	-1.137335	32

Figure 26 cluster labels

- The highlighted labels are the maximum segments.

Companies in each cluster

```
'Coca Cola Company' 'Kansas City Southern' 'Leggett & Platt'
'Lennar Corp.' 'Laboratory Corp. of America Holding' 'LKQ Corporation'
'L-3 Communications Holdings' 'Lilly (Eli) & Co.' 'Lockheed Martin Corp.'
'Alliant Energy Corp' 'Leucadia National Corp.' 'Southwest Airlines'
'Level 3 Communications' 'LyondellBasell' 'Mastercard Inc.'
'Mid-America Apartments' 'Macerich' "Marriott Int'l." 'Masco Corp.'
'Mattel Inc.' "McDonald's Corp." "Moody's Corp" 'Mondelez International'
'MetLife Inc.' 'Mohawk Industries' 'Mead Johnson' 'McCormick & Co.'
'Martin Marietta Materials' 'Marsh & McLennan' '3M Company'
'Monster Beverage' 'Altria Group Inc' 'The Mosaic Company'
'Marathon Petroleum' 'Merck & Co.' 'M&T Bank Corp.' 'Mettler Toledo'
'Mylan N.V.' 'Navient' 'NASDAQ OMX Group' 'NextEra Energy'
'Newmont Mining Corp. (Hldg. Co.)' 'Nielsen Holdings'
'Norfolk Southern Corp.' 'Northern Trust Corp.' 'Nucor Corp.'
'Newell Brands' 'Realty Income Corporation' 'Omnicom Group'
"O'Reilly Automotive" "People's United Financial" 'Pitney-Bowes'
'PACCAR Inc.' 'PG&E Corp.' 'Priceline.com Inc'
'Public Serv. Enterprise Inc.' 'PepsiCo Inc.' 'Pfizer Inc.'
'Principal Financial Group' 'Procter & Gamble' 'Progressive Corp.'
'Pulte Homes Inc.' 'Philip Morris International' 'PNC Financial Services'
'Pentair Ltd.' 'Pinnacle West Capital' 'PPG Industries' 'PPL Corp.'
'Prudential Financial' 'Phillips 66' 'Praxair Inc.' 'PayPal'
'Ryder System' 'Royal Caribbean Cruises Ltd' 'Regeneron'
'Robert Half International' 'Roper Industries' 'Republic Services Inc'
'SCANA Corp' 'Charles Schwab Corporation' 'Spectra Energy Corp.'
'Sealed Air' 'Sherwin-Williams' 'SL Green Realty'
'Scripps Networks Interactive Inc.' 'Southern Co.'
'Simon Property Group Inc' 'S&P Global, Inc.' 'Stericycle Inc'
'Sempra Energy' 'SunTrust Banks' 'State Street Corp.'
'Skyworks Solutions' 'Synchrony Financial' 'Stryker Corp.' 'AT&T Inc'
'Molson Coors Brewing Company' 'Tegna, Inc.' 'Torchmark Corp.'
'Thermo Fisher Scientific' 'TripAdvisor' 'The Travelers Companies Inc.'
'Tractor Supply Company' 'Tyson Foods' 'Tesoro Petroleum Co.'
'Total System Services' 'Texas Instruments' 'Under Armour'
'United Continental Holdings' 'UDR Inc' 'Universal Health Services, Inc.'
'United Health Group Inc.' 'Unum Group' 'Union Pacific'
'United Parcel Service' 'United Technologies' 'Varian Medical Systems'
'Valero Energy' 'Vulcan Materials' 'Vornado Realty Trust'
'Verisk Analytics' 'Verisign Inc.' 'Vertex Pharmaceuticals Inc'
'Ventas Inc' 'Verizon Communications' 'Waters Corporation'
'Wec Energy Group Inc' 'Wells Fargo' 'Whirlpool Corp.'
'Waste Management Inc.' 'Western Union Co' 'Weyerhaeuser Corp.'
'Wyndham Worldwide' 'Xcel Energy Inc' 'XL Capital' 'Exxon Mobil Corp.'
'Dentsply Sirona' 'Xerox Corp.' 'Xylem Inc.' 'Yahoo Inc.'
'Yum! Brands Inc' 'Zimmer Biomet Holdings' 'Zions Bancorp' 'Zoetis']
```

```
In cluster 1, the following companies are present:
['Analog Devices, Inc.' 'Alexion Pharmaceuticals' 'Amazon.com Inc'
 'Apache Corporation' 'Anadarko Petroleum Corp' 'Baker Hughes Inc'
 'Chesapeake Energy' 'Charter Communications' 'Cabot Oil & Gas'
 'Concho Resources' 'Devon Energy Corp.' 'EOG Resources' 'EQT Corporation'
 'Freeport-McMoran Cp & Gld' 'Hess Corporation'
 'Hewlett Packard Enterprise' 'Kinder Morgan' 'Marathon Oil Corp.'
 'Murphy Oil' 'Noble Energy Inc' 'Netflix Inc.' 'Newfield Exploration Co'
 'National Oilwell Varco Inc.' 'ONEOK' 'Occidental Petroleum'
 'Quanta Services Inc.' 'Range Resources Corp.' 'Southwestern Energy'
 'Teradata Corp.' 'Williams Cos.' 'Wynn Resorts Ltd' 'Cimarex Energy']
```

Group by KM_segments & GICS Sector

KM_segments	GICS Sector	Security
0	Consumer Discretionary	37
	Consumer Staples	19
	Energy	8
	Financials	49
	Health Care	39
	Industrials	52
	Information Technology	29
	Materials	19
	Real Estate	27
	Telecommunications Services	5
	Utilities	24
1	Consumer Discretionary	3
	Energy	22
	Health Care	1
	Industrials	1
	Information Technology	4
	Materials	1

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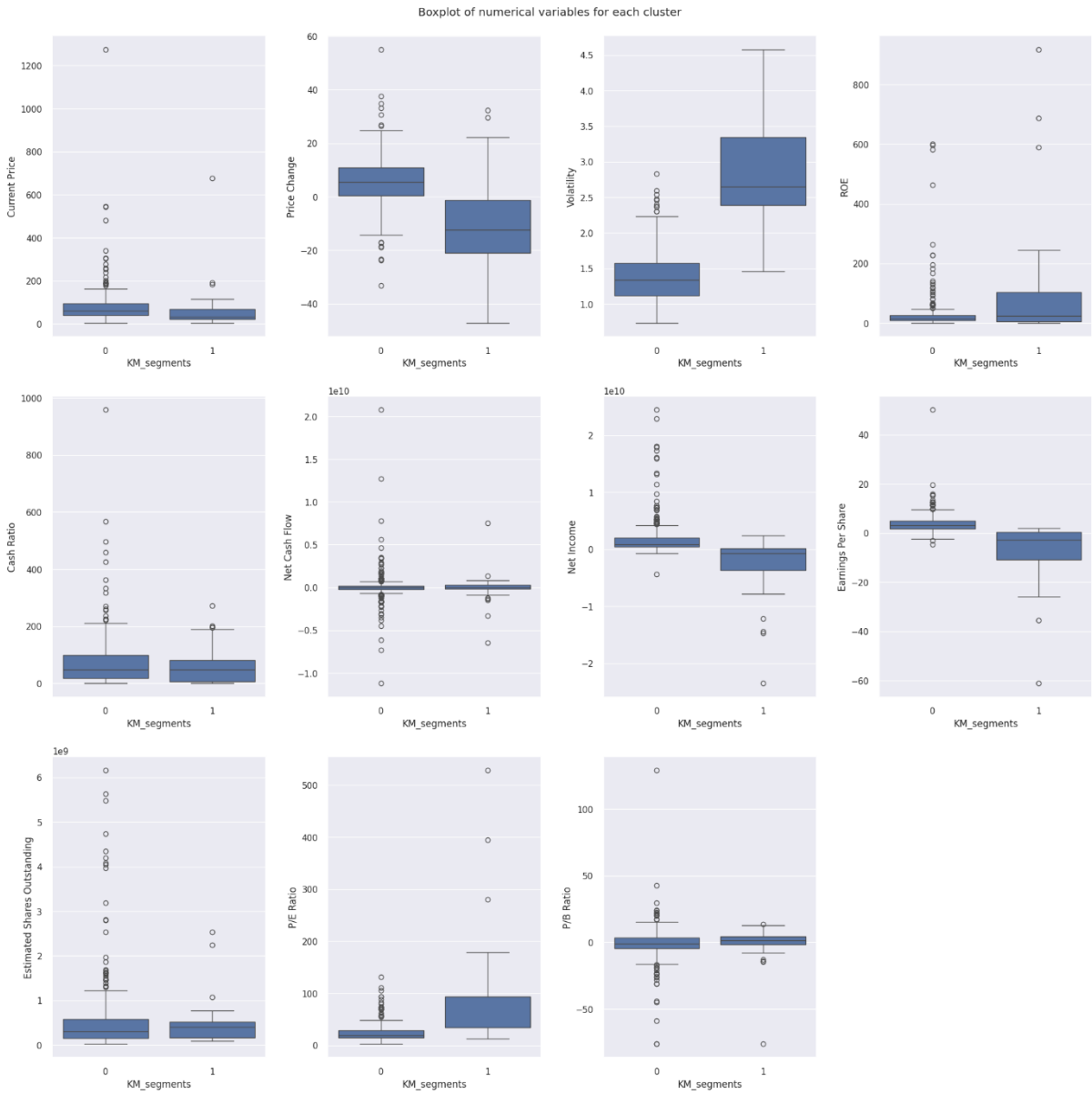
Table 5 Group by KM_segments & GICS Sector

Selecting numerical columns



8.7 Insights:-

Cluster 0

- The current price in this cluster is more than the other cluster.
- The price change is higher than the other cluster.
- Volatility is lower in this cluster compared to cluster 1.
- ROE is lower than the other cluster.
- The cash ratio is greater than that of Cluster 1, with a moderate mean.
- Net cash flow is moderate.
- Net income is higher than cluster 1.

- Earnings per share is better than other cluster.
- Estimated shares outstanding are lower in this cluster.
- The P/E Ratio is lower in this cluster.
- The P/B Ratio is moderate in both cluster 0 & cluster 1.

Cluster 1

- The current price in Cluster 1 is less than Cluster 0.
- Price change in this cluster is lower than Cluster 0 and Cluster 1 has only two outliers.
- Volatility in Cluster 1 is more than Cluster 0 and it has no outliers.
- ROE in Cluster 1 is higher compared to Cluster 0.
- The cash ratio in this cluster is more than that of cluster 0.
- Net Cash Flow is moderate in both Cluster 0 and Cluster 1.
- The net income in this cluster is lower than that of Cluster 0.
- Earnings per share in Cluster 1 is less than Cluster 0.
- The estimated shares outstanding for this cluster are not more than cluster 0.
- P/E Ratio in Cluster 1 is more than in Cluster 0.
- The P/B Ratio in Cluster 1 is better than Cluster 1.

9. This is formatted as code

Hierarchical Clustering: –

9.1 Computing Cophenetic Correlation

```
Cophenetic correlation for Euclidean distance and single linkage is 0.9232271494002922.
Cophenetic correlation for Euclidean distance and complete linkage is 0.7873280186580672.
Cophenetic correlation for Euclidean distance and average linkage is 0.9422540609560814.
Cophenetic correlation for Euclidean distance and weighted linkage is 0.8693784298129404.
Cophenetic correlation for Chebyshev distance and single linkage is 0.9062538164750717.
Cophenetic correlation for Chebyshev distance and complete linkage is 0.598891419111242.
Cophenetic correlation for Chebyshev distance and average linkage is 0.9338265528030499.
Cophenetic correlation for Chebyshev distance and weighted linkage is 0.9127355892367.
Cophenetic correlation for Mahalanobis distance and single linkage is 0.925919553052459.
Cophenetic correlation for Mahalanobis distance and complete linkage is 0.7925307202850002.
Cophenetic correlation for Mahalanobis distance and average linkage is 0.9247324030159736.
Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.8708317490180428.
Cophenetic correlation for Cityblock distance and single linkage is 0.9334186366528574.
Cophenetic correlation for Cityblock distance and complete linkage is 0.7375328863205818.
Cophenetic correlation for Cityblock distance and average linkage is 0.9302145048594667.
Cophenetic correlation for Cityblock distance and weighted linkage is 0.731045513520281.
**********************************************************************************************
Highest cophenetic correlation is 0.9422540609560814, which is obtained with Euclidean distance and average linkage.
```

Figure 27 Cophenetic Correlation

Let's explore different linkage methods with Euclidean distance only.

combination of distance metric and linkage method with the highest cophenetic correlation

```
Cophenetic correlation for single linkage is 0.9232271494002922.
Cophenetic correlation for complete linkage is 0.7873280186580672.
Cophenetic correlation for average linkage is 0.9422540609560814.
Cophenetic correlation for ward linkage is 0.7101180299865353.
Cophenetic correlation for centroid linkage is 0.9314012446828154.
Cophenetic correlation for median linkage is 0.9198690668829905.
**********************************************************************************************
Highest cophenetic correlation is 0.9422540609560814, which is obtained with average linkage.
```

Figure 28 combination of distance metric and linkage method with the highest cophenetic correlation

Let's view the dendrograms for the different linkage methods with Euclidean distance.

9.2 Checking Dendrograms

list of linkage methods

linkage_methods are single', 'complete', 'average', 'ward', 'centroid', 'median'

We will enumerate through the list of linkage methods above

For each linkage method, we will plot the dendrogram and calculate the cophenetic correlation

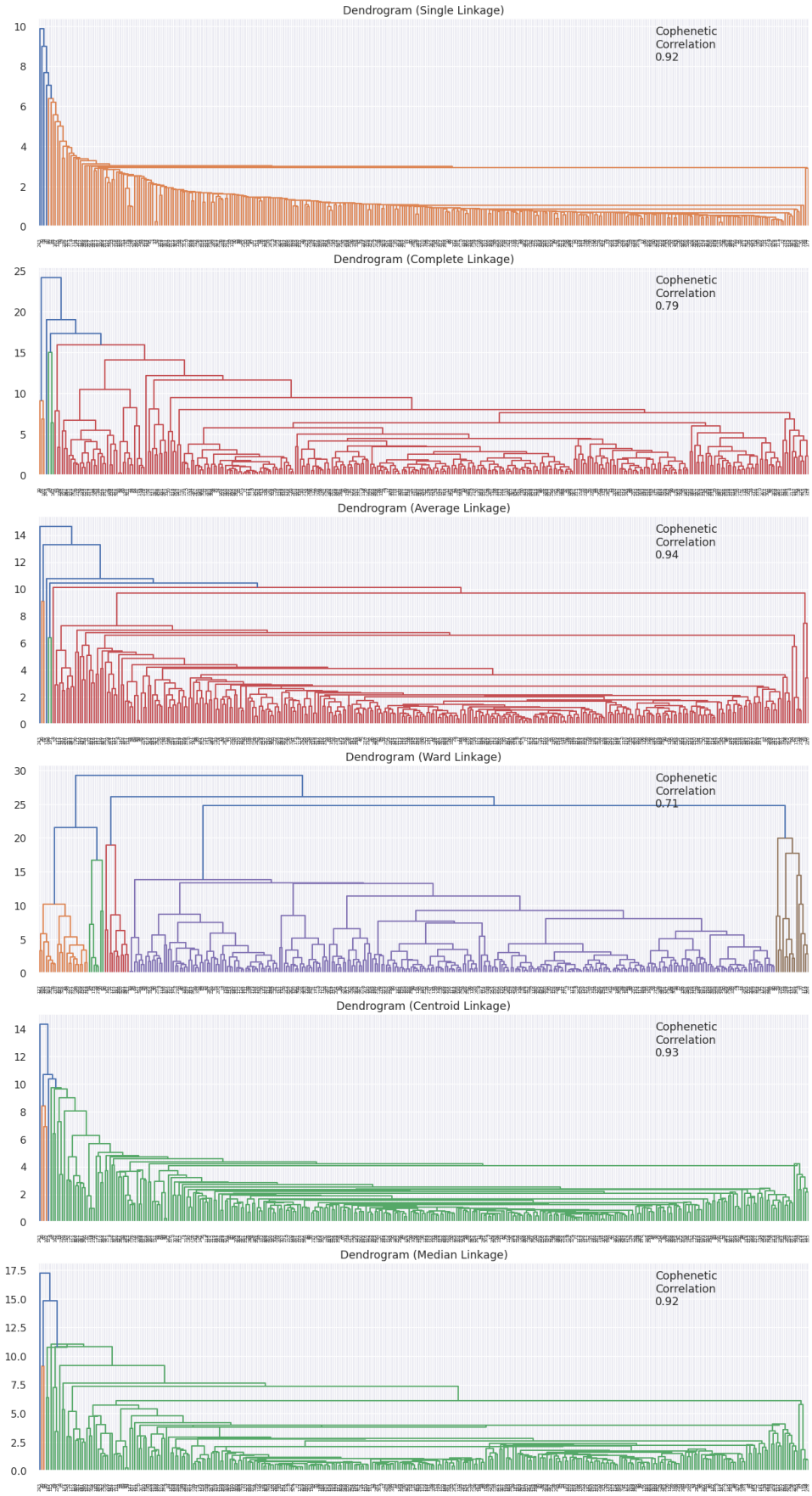


Figure 29 Checking Dendrograms

Observations: -

- Looking the above dendrograms, the Ward linkage seems to result in the best separation between clusters, even though its cophenetic correlation is lower than the other linkages.
- 4 looks to be a good choice for no. of clusters.

Dataframe to compare cophenetic correlations for different linkage methods

	Linkage	Cophenetic Coefficient
3	ward	0.710118
1	complete	0.787328
5	median	0.919869
0	single	0.923227
4	centroid	0.931401
2	average	0.942254

Table 6 cophenetic correlations for different linkage methods

9.3 Creating model using sklearn

```
AgglomerativeClustering
AgglomerativeClustering(linkage='average', n_clusters=6)
```

9.4 Cluster Profiling

Group by the cluster labels reflected in the below table.

HC_segments	Current Price	Price Change	Volatility	ROE	Cash Ratio	Net Cash Flow	Net Income	Earnings Per Share	Estimated Shares Outstanding	P/E Ratio	P/B Ratio	count_in_each_segment
0	77.287589	4.099730	1.518066	35.336336	66.900901	-33197321.321321	1538074666.666667	2.885270	560505037.293544	32.441706	-2.174921	333
1	25.640000	11.237908	1.322355	12.500000	130.500000	16755500000.000000	13654000000.000000	3.295000	2791829362.100000	13.649696	1.508484	2
2	24.485001	-13.351992	3.482611	802.000000	51.000000	-1292500000.000000	-19106500000.000000	-41.815000	519573983.250000	60.748608	1.565141	2
3	104.660004	16.224320	1.320606	8.000000	958.000000	592000000.000000	3669000000.000000	1.310000	2800763359.000000	79.893133	5.884467	1
4	1274.949951	3.190527	1.268340	29.000000	184.000000	-1671386000.000000	2551360000.000000	50.090000	50935516.070000	25.453183	-1.052429	1
5	276.570007	6.189286	1.116976	30.000000	25.000000	90885000.000000	596541000.000000	8.910000	66951851.850000	31.040405	129.064585	1

Table 7 Cluster Profiling

HC _ Companies in each cluster

```
'Mead Johnson' 'McCormick & Co.' 'Martin Marietta Materials'
'Marsh & McLennan' '3M Company' 'Monster Beverage' 'Altria Group Inc'
'The Mosaic Company' 'Marathon Petroleum' 'Merck & Co.'
'Marathon Oil Corp.' 'M&T Bank Corp.' 'Mettler Toledo' 'Murphy Oil'
'Mylan N.V.' 'Navient' 'Noble Energy Inc' 'NASDAQ OMX Group'
'NextEra Energy' 'Newmont Mining Corp. (Hldg. Co.)' 'Netflix Inc.'
'Newfield Exploration Co' 'Nielsen Holdings'
'National Oilwell Varco Inc.' 'Norfolk Southern Corp.'
'Northern Trust Corp.' 'Nucor Corp.' 'Newell Brands'
'Realty Income Corporation' 'ONEOK' 'Omnicom Group' "O'Reilly Automotive"
'Occidental Petroleum' "People's United Financial" 'Pitney-Bowes'
'PACCAR Inc.' 'PG&E Corp.' 'Public Serv. Enterprise Inc.' 'PepsiCo Inc.'
'Pfizer Inc.' 'Principal Financial Group' 'Procter & Gamble'
'Progressive Corp.' 'Pulte Homes Inc.' 'Philip Morris International'
'PNC Financial Services' 'Pentair Ltd.' 'Pinnacle West Capital'
'PPG Industries' 'PPL Corp.' 'Prudential Financial' 'Phillips 66'
'Quanta Services Inc.' 'Praxair Inc.' 'PayPal' 'Ryder System'
'Royal Caribbean Cruises Ltd' 'Regeneron' 'Robert Half International'
'Roper Industries' 'Range Resources Corp.' 'Republic Services Inc'
'SCANA Corp' 'Charles Schwab Corporation' 'Spectra Energy Corp.'
'Sealed Air' 'Sherwin-Williams' 'SL Green Realty'
'Scripps Networks Interactive Inc.' 'Southern Co.'
'Simon Property Group Inc' 'S&P Global, Inc.' 'Stericycle Inc'
'Sempra Energy' 'SunTrust Banks' 'State Street Corp.'
'Skyworks Solutions' 'Southwestern Energy' 'Synchrony Financial'
'Stryker Corp.' 'AT&T Inc' 'Molson Coors Brewing Company'
'Teradata Corp.' 'Tegna, Inc.' 'Torchmark Corp.'
'Thermo Fisher Scientific' 'TripAdvisor' 'The Travelers Companies Inc.'
'Tractor Supply Company' 'Tyson Foods' 'Tesoro Petroleum Co.'
'Total System Services' 'Texas Instruments' 'Under Armour'
'United Continental Holdings' 'UDR Inc' 'Universal Health Services, Inc.'
'United Health Group Inc.' 'Unum Group' 'Union Pacific'
'United Parcel Service' 'United Technologies' 'Varian Medical Systems'
'Valero Energy' 'Vulcan Materials' 'Vornado Realty Trust'
'Verisk Analytics' 'Verisign Inc.' 'Vertex Pharmaceuticals Inc'
'Ventas Inc' 'Verizon Communications' 'Waters Corporation'
'Wec Energy Group Inc' 'Wells Fargo' 'Whirlpool Corp.'
'Waste Management Inc.' 'Williams Cos.' 'Western Union Co'
'Weyerhaeuser Corp.' 'Wyndham Worldwide' 'Wynn Resorts Ltd'
'Cimarex Energy' 'Xcel Energy Inc' 'XL Capital' 'Exxon Mobil Corp.'
'Dentsply Sirona' 'Xerox Corp.' 'Xylem Inc.' 'Yahoo Inc.'
'Yum! Brands Inc' 'Zimmer Biomet Holdings' 'Zions Bancorp' 'Zoetis']
```

Figure 30 HC _ Companies in each cluster

```
In cluster 5, the following companies are present:
['Alliance Data Systems']

In cluster 2, the following companies are present:
['Apache Corporation' 'Chesapeake Energy']

In cluster 1, the following companies are present:
['Bank of America Corp' 'Intel Corp.']

In cluster 3, the following companies are present:
['Facebook']

In cluster 4, the following companies are present:
['Priceline.com Inc']
```

Figure 31 HC _ Cluster wise companies

Group by the HC_segments & GICS Sector

HC_segments	GICS Sector	Security
0	Consumer Discretionary	39
	Consumer Staples	19
	Energy	28
	Financials	48
	Health Care	40
	Industrials	53
	Information Technology	30
	Materials	20
	Real Estate	27
	Telecommunications Services	5
	Utilities	24
1	Financials	1
	Information Technology	1
2	Energy	2
3	Information Technology	1
4	Consumer Discretionary	1
5	Information Technology	1

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Table 8 Group by the HC_segments & GICS Sector

Boxplot of numerical variables for each cluster

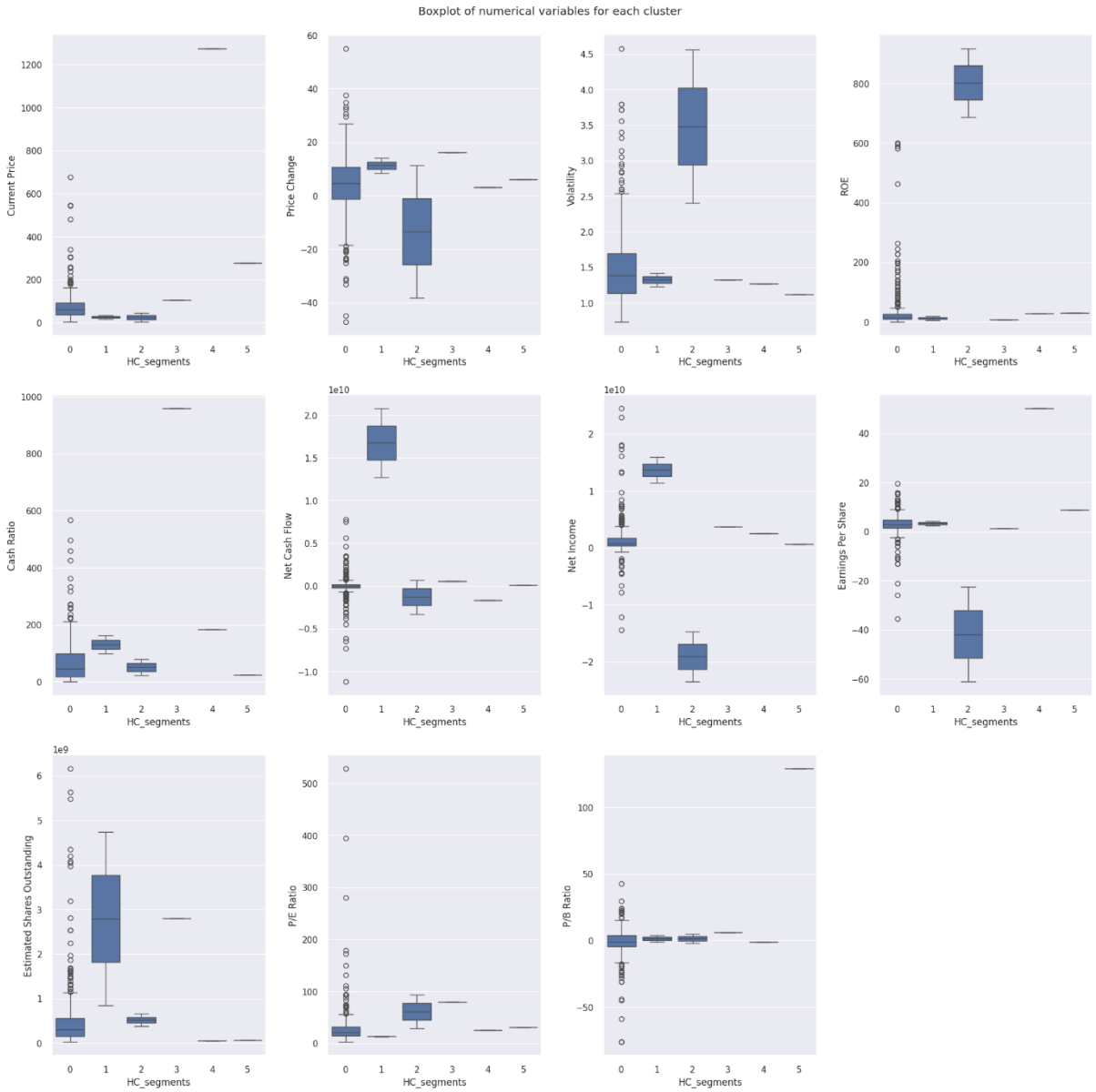

Figure 32 Boxplot of numerical variables for each cluster

9.5 K-means vs Hierarchical Clustering

You compare several things, like:

- Which clustering technique took less time for execution?
  - Ans**:** Distortion Score Elbow for the KMeans Clustering technique took less time for execution/fit time (i.e between 0.004 to 0.012 seconds) compared to Silhouette Score Elbow for KMeans Clustering technique (fit time from 0.005 seconds to 0.025 seconds).
- Which clustering technique gave you more distinct clusters, or are they the same?

  Ans: Hierarchical clusters more distinct clusters 6 compared to KMeans cluster 2.

  Hierarchical clusters are 5 but they are not the same.

  KMeans cluster is 1.

- How many observations are there in the similar clusters of both algorithms?

  Ans: There are no similar clusters in both algorithms so no observations are the same.

- How many clusters are obtained as the appropriate number of clusters from both algorithms?

  Ans: KMeans obtained cluster 1 and Hierarchical clusters obtained cluster 3, cluster 4, and cluster 5 as the appropriate numbers from both algorithms. A total of 4 clusters was obtained from both clusters.

  KMeans cluster 1 and Hierarchical cluster 2 have a combined company 'Apache Corporation'.

You can also mention any differences or similarities you obtained in the cluster profiles from both clustering techniques.

- Segments prepared for based on numerical variable for each cluster of both K-Means and Hierarchical clusters.

- Although some of the countries have been swapped between K-means and Hierarchical clusters, we can deduce that overall for both the methodologies, the characteristics of the clusters are similar.
- A few other countries have swapped clusters between K-means and Hierarchical clustering.

10. Actionable Insights and Recommendations

10.1 Insights

- The current price has many outliers so the price is very high and unpredictable.
- Stock prices have more volatility.
- Telecommunication services and Materials contributed less business operations.
- Sub-industry wise also showing less portion of services.
- ROE has a strong adverse correlation (i.e -0.54) with Earnings Per Share.
- K-Means Cluster gives the appropriate one based on the highest scores which are less accurate.
- Silhouette coefficient values are sometimes below the average scoreline.
- Some Clustering methods have more Clusters which increases the complexity.

10.2 Recommendations

- It needs to reduce Outliers to maintain the consistency of the current price.
- Volatility needs to be controlled by choosing the right stocks having less volatility.
- Require more support to expand the sector-wise business operations for telecommunication services and Materials.
- Sub-industry has to increase production to improve business.
- Earnings per share have to increase by raising the share price.
- K-Means Cluster should maintain a fixed group of observations.
- Silhouette coefficient values need to be above the average.
- Always do easy clusters with fewer groups to avoid confusion, complexity, and time saving.